# RAW ESSENTIALS

# SCOPE OF WORK

## MVP Phase 0 Specification

Blockchain-Based B2B Trade Finance Platform
Gold & Cash Backed Tokenized Financing | Dubai, UAE

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|---|---|
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| **Author** | Dinesh - Technical Lead / Sr. Blockchain Solution Architect |
| **Reviewed By** | |

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# 1. Executive Summary

## 1.1 Market Opportunity

Raw Essentials is building a blockchain-based B2B trade finance platform on Avalanche Evergreen Subnet targeting Dubai's supply chain market. The platform operates a Platform-as-Financier model, tokenizing physical gold and cash assets to create liquidity pools that finance trade deals across Food, Energy, and Vehicle verticals.

This addresses a critical market gap: the global trade finance gap reached an estimated $2.5 trillion by 2025, with the UAE segment alone exceeding $3.3 billion (ICC 2020). SMEs face a 45% rejection rate on trade finance applications globally, rising to 50% or higher in the MENA region. UAE SMEs represent 94% of all enterprises, contribute 86% of private sector employment and over 60% of non-oil GDP, yet receive only 9.5% of total bank lending (Central Bank of UAE, H1 2024). Suppliers face 60-120 day payment delays, strangling working capital and limiting growth.

Raw Essentials solves this by enabling trade financing approval in under 20 minutes (versus 2-4 weeks traditional), backed by real physical assets (gold and cash), with full regulatory compliance through Dubai's VARA framework. The platform operates as a closed-loop permissioned environment where every participant is KYC-verified and every transaction is auditable.

## 1.2 Platform Overview

**KEY NUMBERS**

Target Market: $3.3B UAE trade finance gap | 94% of UAE enterprises are SMEs

Pilot Program: 3 deals totaling $16M across Food ($5M), Energy ($8M), Vehicles ($3M)

Speed: < 20 minute financing approval vs. 2-4 week traditional bank process

Providers: Minimum 4 asset providers (2 Gold, 2 Cash) with $10M+ liquidity pool

Project Value: $2,400,000 | Timeline: 5 Phases across approximately 12-15 months

Revenue Model: 3-5% financing fee per trade, split 70% providers / 20% platform / 10% reserve

## 1.4 How the Platform Works (Simplified)

STEP 1 - PROVIDERS DEPOSIT: Gold providers deposit physical gold (verified by custodians like Brinks/Loomis) and receive TCT tokens representing their gold. Cash providers wire funds and receive CCT tokens. Both receive LP tokens reflecting their share of the unified liquidity pool.

STEP 2 - POOL FORMATION: TCT and CCT tokens form a unified liquidity pool valued in USD equivalent. Gold-backed tokens are valued using LBMA price feeds. The pool represents the total financing capacity available to traders.

STEP 3 - TRADE FINANCING: SME traders (buyers) apply for financing. AI-assisted document verification and credit scoring evaluate the application. Approved trades are funded from the pool and locked in smart contract escrow with milestone-based release (30/40/30 split).

STEP 4 - SETTLEMENT: Suppliers receive milestone payments as goods are delivered and verified. Suppliers convert tokens to fiat (AED/USD) via Chill Pay off-ramp. Buyer repays financing over agreed terms. Financing fees are distributed: 70% to providers as yield, 20% platform revenue, 5% reserve fund, 5% operations.

STEP 5 - YIELD: Providers earn 6-8% target APY from financing fees, distributed proportional to their LP share. Providers can withdraw at any time subject to liquidity availability.

# 2. Product Objectives & Success Criteria

## 2.1 Business Objectives

| ID | Objective | Description & Rationale |
|---|---|---|
| **BO-1** | **Validate Platform Model** | Execute 3 pilot trade deals totaling $16M across Food ($5M), Energy ($8M), and Vehicles ($3M) to prove the tokenized trade finance model works with real counterparties and real money in a regulated environment. |
| **BO-2** | **Rapid Trade Financing** | Achieve trade financing approval in under 20 minutes versus 2-4 weeks with traditional banks, demonstrating 100x speed improvement through blockchain-based automation. |
| **BO-3** | **Asset Provider Ecosystem** | Onboard minimum 2 Gold Providers and 2 Cash Providers to establish a multi-asset liquidity pool exceeding $10M, proving the dual-asset tokenization model attracts institutional capital. |
| **BO-4** | **Regulatory Readiness** | Build platform fully compliant with VARA framework (Law No. 4 of 2022): permissioned blockchain, KYC-gated access, audit trails, regulator observer capability, GOAML reporting integration. |
| **BO-5** | **100% Asset Backing** | Maintain 100%+ backing ratio for all tokens at all times with real-time Proof of Reserves dashboard, meeting VARA's ARVA (Asset-Referenced Virtual Asset) requirements. |
| **BO-6** | **Operational Excellence** | Deliver platform with 99.5% uptime SLA, zero critical smart contract vulnerabilities post-audit, complete audit trail for every transaction, and sub-2-second on-chain finality. |

## 2.2 Measurable Success Criteria

| ID | Criterion | Target | Measurement Method |
|---|---|---|---|
| **SC-1** | Pilot Deals Completed | 3 deals / $16M | Trade completion records + on-chain settlement confirmation |
| **SC-2** | Financing Approval Time | < 20 min (P90) | Workflow timestamp delta: application submitted to approval |
| **SC-3** | Backing Ratio | >= 100% always | Real-time dashboard: on-chain token supply vs verified off-chain reserves |
| **SC-4** | Platform Uptime | 99.5% SLA | Grafana/Prometheus monitoring, monthly uptime calculation |
| **SC-5** | Smart Contract Audit | 0 Critical, 0 High | Third-party audit report (CertiK, Trail of Bits, or equivalent) |

| | | | |
|---|---|---|---|
| **SC-6** | Asset Providers | >= 4 (2G + 2C) | Completed KYC + active deposit + receiving yield distributions |
| **SC-7** | API Response Time | < 200ms (P95) | APM monitoring (Datadog/NewRelic), excluding on-chain operations |
| **SC-8** | On-Chain Finality | < 2 seconds | Block explorer metrics + event listener latency measurements |

## 2.3 Assumptions & Constraints

Every assumption below has a documented fallback. If any assumption proves false, the fallback is activated without requiring a full re-scope. The Engineering Manager and Product Owner must review these monthly.

### Business Assumptions

| ID | Assumption | Impact if False / Fallback Plan |
|---|---|---|
| **A-1** | VARA license obtained before production launch | Platform cannot operate commercially in Dubai. FALLBACK: Operate in DIFC Innovation Testing License (sandbox) with limited transaction volumes until full license obtained. |
| **A-2** | Minimum 2 Gold + 2 Cash providers committed for pilot | Insufficient liquidity for $16M pilot deals. FALLBACK: Start with single-asset pool (cash-only), offer enhanced yield incentives (8-10% APY) to attract first providers. |
| **A-3** | 3 pilot trade deals pre-arranged by Gary (Food $5M, Energy $8M, Vehicles $3M) | No real transaction data to validate platform. FALLBACK: Execute simulated deals with testnet tokens using real counterparty data for process validation. |
| **A-4** | Gold custodian (Brinks or Loomis) cooperates with verification process | Cannot verify physical gold backing digitally. FALLBACK: Manual email/document verification workflow for MVP (already planned). API integration deferred to Phase 1. |
| **A-5** | Chill Pay partnership signed with API access for fiat off-ramp | Suppliers cannot convert tokens to fiat through platform. FALLBACK: Manual wire transfer via admin console + parallel evaluation of alternate off-ramp providers (Wyre, MoonPay). |
| **A-6** | Full engineering team available by Phase 1 kickoff | Development delays and extended timeline. FALLBACK: Sequential module development with reduced parallelism; critical path modules first. |
| **A-7** | AWS Bahrain region (me-south-1) meets latency/performance requirements | Infrastructure performance issues. FALLBACK: AWS Dubai Edge location or multi-region deployment with me-south-1 as primary. |
| **A-8** | Shufti Pro supports UAE entity verification (trade license, UBO) | KYC provider cannot handle UAE-specific documents. FALLBACK: Sumsub or Jumio as alternative. Manual KYC review as interim. |

### Technical Constraints

| ID | Constraint | Rationale |
|---|---|---|
| **TC-1** | UAE data residency - AWS me-south-1 (Bahrain) | VARA regulations and UAE PDPL (Federal Decree-Law No. 45/2021) require personal data to remain within UAE/GCC region. |
| **TC-2** | Permissioned blockchain only (Avalanche Evergreen Subnet) | VARA KYC requirements mandate that all blockchain participants are verified. Public permissionless chains cannot satisfy this. |
| **TC-3** | Zero PII stored on-chain | GDPR/PDPL right-to-deletion is incompatible with blockchain immutability. Only hashes and references stored on-chain. |
| **TC-4** | HSM-managed keys (AWS CloudHSM FIPS 140-2 Level 3) | Institutional-grade custody requirement for asset-backed tokens handling millions in value. Keys never leave HSM boundary. |
| **TC-5** | AED primary settlement currency, USD secondary | UAE domestic market requirement. International suppliers may elect USD settlement via Chill Pay. |
| **TC-6** | 48-hour upgrade timelock on smart contracts | Regulatory transparency requirement. All contract upgrades visible for 48 hours before execution. Emergency 3-of-5 multisig bypass available. |

| | | |
|---|---|---|
| **TC-7** | Maximum 25% pool allocation per single trade | Concentration risk limit protecting provider capital. No single deal can consume more than one-quarter of total pool. |
| **TC-8** | Minimum 100% backing ratio enforced on-chain | VARA ARVA framework requires full backing for all asset-referenced virtual assets at all times. |

## 2.4 Risk Register

Risk scoring: Likelihood (1-5) x Impact (1-5) = Risk Score. Scores 12+ are CRITICAL and require active mitigation with named owner. Scores 8-11 are HIGH and require documented mitigation. This register is reviewed bi-weekly during development.

| ID | Risk Description | L | I | S | Mitigation Strategy | Owner |
|---|---|---|---|---|---|---|
| **R-1** | VARA license delayed beyond Month 6 | 3 | 5 | **15** | Engage VARA pre-application immediately. DIFC Innovation Testing License as interim. Retained legal counsel specializing in VA regulation. | Gary / Legal |
| **R-2** | Gold custodian refuses API integration | 4 | 3 | **12** | MVP already designed for manual email verification. API integration is Phase 1 enhancement, not MVP blocker. | Dinesh / Ops |
| **R-3** | Chill Pay partnership not finalized | 3 | 4 | **12** | Manual wire transfer fallback ready. Parallel evaluation of Wyre, MoonPay, and TransFi as alternative off-ramp providers. | Gary / Dinesh |
| **R-4** | Trader default on pilot deal ($5-8M exposure) | 2 | 5 | **10** | Reserve fund (5% of all fees). Milestone-based release (30/40/30) limits exposure. Credit scoring gates access. Trade credit insurance exploration. | Dinesh / Risk |
| **R-5** | Smart contract vulnerability post-deployment | 2 | 5 | **10** | Third-party audit (CertiK). >90% test coverage. Emergency pause capability. Timelocked upgrades. Bug bounty program post-launch. | Tech Lead |
| **R-6** | Oracle price manipulation / stale data | 2 | 5 | **10** | Multi-source LBMA feeds. Circuit breaker at 5% deviation from median. Manual admin override. 30-minute maximum staleness threshold. | Blockchain Team |
| **R-7** | Scope creep during development | 4 | 3 | **12** | SOW v4.0 as frozen baseline. Formal change control process (Change Request Form + impact assessment + sign-off). Sprint scope locked after planning. | PM / EM |

| | | | | | | |
|---|---|---|---|---|---|---|
| **R-8** | Insufficient provider liquidity for pilot deals | 3 | 3 | **9** | Pre-arranged commitments from Gary's network. Flexible pool structure. Enhanced yield incentives for early providers. Phased pilot (smaller deal first). | Gary / BD |
| **R-9** | Key personnel departure mid-project | 2 | 4 | **8** | Documentation-first approach. Mandatory code reviews. Weekly knowledge sharing sessions. No single-person dependencies on critical modules. | EM |
| **R-10** | Regulatory requirement changes mid-build | 2 | 4 | **8** | Modular compliance layer (easily swappable). Retained regulatory counsel monitoring VARA updates. Architecture designed for flexibility. | Legal / Dinesh |

# 3. User Personas & Journeys

## 3.1 User Personas

| Persona | Profile | Goals | Pain Points Addressed |
|---|---|---|---|
| **Gold Provider** | Institutional entity with physical gold: vaults, bullion banks, family offices, refineries | Earn 6-8% APY on idle gold. Transparent yield tracking. Flexible withdrawal. Full regulatory compliance. Real-time portfolio view. | Gold sitting at 0% return. No access to trade finance yields. Opaque fund management. Long lock-in periods. Complex reporting. |
| **Cash Provider** | HNWIs, family offices, corporate treasuries with idle USD/AED capital | Earn 6-8% APY vs 1-3% bank deposits. Source of funds compliance. Flexible entry/exit. Transparent risk metrics. | Low bank deposit rates. Limited trade finance access. Liquidity concerns with alternatives. Complex onboarding. Opaque risk. |
| **Trader (Buyer)** | SME importers in Food, Energy, Vehicle verticals. UAE/GCC based. | Rapid financing (< 20 min). Competitive rates (3-5%). Digital workflow. Credit history building. Milestone-based flexibility. | 70%+ bank rejection. 2-4 week approvals. Excessive documentation. No credit recognition from prior trades. Cash flow gaps. |
| **Trader (Supplier)** | Exporters from source countries: Thailand, India, Brazil, UAE domestic | Fast settlement. Milestone releases (30/40/30). Reduced counterparty risk. Digital documents. Fiat off-ramp. | 60-120 day payment delays. Cash flow constraints. Document fraud risk. Settlement disputes. Currency conversion complexity. |
| **Platform Admin** | Internal operations: compliance officers, risk managers, technical support, super admins | Efficient verification. Compliance reporting (GOAML, VARA). System monitoring. Risk management. Dispute resolution. | Manual reconciliation. Fragmented tools. Audit complexity. Alert fatigue. No unified dashboard. Regulatory reporting burden. |
| **Regulator Observer** | VARA compliance auditors with read-only platform access | Real-time oversight. Transaction monitoring. Compliance verification. Audit trail access. Reporting validation. | Delayed reporting from VASPs. Incomplete audit trails. Manual data requests. Limited real-time visibility. |

## 3.2 Primary User Journeys

### Journey 1: Gold Provider - Deposit to Yield

| Step | Action | System Behavior | Outcome |
|---|---|---|---|
| 1 | Gold Provider registers and completes KYC | Provider Portal: Self-service registration, entity type selection, email verification. Shufti Pro KYC: ID verification, trade license upload, UBO declaration, sanctions screening. On-chain: DIDVerifier.setVerified(true). | Provider account active. On-chain verified status. Can access Provider Portal. |
| 2 | Provider submits gold deposit | Enter: weight (troy oz), purity (fineness), custodian name, vault location, insurance details. Upload 4 documents: Certificate of Ownership, | Deposit request created. Status: PENDING_VERIFICATION. |

| | | | |
|---|---|---|---|
| | | Assay Report (LBMA-certified), Insurance Certificate, Custodian Confirmation Letter. | |
| 3 | Platform verifies gold custody | Admin Console: Compliance officer reviews all 4 documents. MVP: Manual email to custodian requesting vault confirmation. Custodian responds with signed confirmation. Admin approves or rejects (rejection requires written reason). | Status: VERIFIED or REJECTED. Provider notified via email + portal. |
| 4 | On-chain registration and tokenization | AssetTokenFactory.registerGold() stores metadata hash on-chain (weight, purity, custodian, vault - no PII). LiquidityPool.deposit() mints TCT tokens (1 TCT = 1 troy oz gold equivalent at LBMA spot). LP tokens issued proportional to USD-equivalent pool share. | TCT tokens in provider wallet. LP tokens reflecting pool share. Proof of Reserves dashboard updated. |
| 5 | Yield accrual begins | YieldEngine runs daily calculation: trade financing fees collected, 70% allocated to provider pool, distributed proportional to LP share. Provider Portal dashboard shows: total deposited, current gold price, LP share %, accumulated yield, projected APY. | Provider earns daily yield. Target: 6-8% APY. Visible in real-time. |

## Journey 2: Trade Financing - Application to Settlement

| Step | Action | System Behavior | Outcome |
|---|---|---|---|
| 1 | Buyer applies for trade financing | Trader Portal: Upload trade documents (SBLC, ICPO, SCO, LOI, commercial invoices, proforma invoices, shipping documents). Enter deal details: commodity type, quantity, value (USD/AED), counterparty (supplier), proposed milestones, repayment terms. | Application created. Status: PENDING_REVIEW. Unique Trade ID assigned. |
| 2 | AI verification + Credit scoring | InvoiceNFT.validateInvoice() runs AI document verification (authenticity, consistency, completeness). CreditScore.getScore() calculates risk. MVP APPROACH: Rules-based scoring with manual admin override (addresses cold-start - no historical data for 3 pilot deals). Scoring factors: trade value, commodity type, counterparty history, documentation quality. | Credit score assigned. Credit limit calculated. Admin review queue populated. |
| 3 | Admin approval and deal structuring | Admin reviews: credit score, document verification results, counterparty data, pool capacity (must not exceed 25% concentration limit). Admin approves deal with terms: financing amount, fee rate (3-5%), milestone structure (default 30/40/30), repayment schedule. | Application APPROVED. Terms finalized. Both parties notified. |
| 4 | Trade NFT minted + Escrow funded | TradeNFT minted with deal metadata (value, parties, milestones, terms). MilestonePayment.sol creates 3 milestones (30/40/30). | Funds locked in smart contract escrow. Milestones created on-chain. Pool capacity reduced. |

| | | | |
|---|---|---|---|
| | | EscrowVault.lockFunds() transfers financing amount from liquidity pool to escrow. Pool utilization updated. | |
| 5 | Milestone verification + release | For each milestone: Supplier ships goods and uploads proof (shipping docs, inspection reports). Buyer confirms receipt. Admin verifies documentation completeness. MilestonePayment transitions: PENDING > APPROVED > RELEASED. EscrowVault releases milestone amount to supplier. | Supplier receives 30%, then 40%, then 30% as milestones complete. |
| 6 | Settlement and off-ramp | Supplier converts tokens via Chill Pay integration (redirect to Chill Pay for AED/USD conversion, T+1 settlement). Platform fee (3-5%) already deducted from financing. Buyer repayment tracked by RepaymentTracker. Trade NFT status: COMPLETED. | Trade settled. Yield distributed to providers. Pool replenished from repayment. |

## Journey 3: Cash Provider - Deposit to Withdrawal

| Step | Action | System Behavior | Outcome |
|---|---|---|---|
| 1 | Cash Provider initiates deposit | Provider Portal: Enter deposit amount (USD/AED), upload Source of Funds documentation (bank statement, wealth declaration). Platform displays wire instructions: bank name, account number, SWIFT/IBAN, reference number. | Deposit request created. Status: AWAITING_WIRE. Wire reference generated. |
| 2 | Wire received and verified | Open Finance Banking API confirms wire receipt (matched by reference number). AML screening runs (GOAML check, sanctions screening). Compliance officer reviews source of funds. CCT tokens minted: 1 CCT = 1 USD. LP tokens issued proportional to pool share. | CCT tokens in wallet. LP share updated. Pool capacity increased. Status: ACTIVE. |
| 3 | Provider requests withdrawal | Full or partial withdrawal. System checks available liquidity (total pool minus locked-in-escrow). If sufficient: immediate processing, T+1 wire. If insufficient: FIFO withdrawal queue with estimated wait time displayed. Yield calculated up to withdrawal date. LP tokens + CCT burned. | Fiat wire initiated (T+1). Tokens burned. LP share reduced. Pool capacity decreased. |

## Journey 4: Trade Dispute Resolution

Disputes are inevitable in a $16M pilot program involving physical goods, cross-border suppliers, and milestone-based payments. This journey defines the complete dispute lifecycle: who can dispute, when, what evidence is required, who arbitrates, and what outcomes are possible. Without this framework, teams building the milestone payment and escrow modules would need to invent their own rules.

### Phase A: Dispute Initiation

| Step | Action | System Behavior | Outcome |
|---|---|---|---|
| 1 | Party raises dispute | Buyer, Supplier, or Platform Admin raises dispute on a specific milestone via their respective portal. Must select dispute category: Quality Deficiency, Documentation Mismatch, Delivery Shortfall, Specification Non-Conformance, Late Delivery, Damaged Goods, Other. | MilestonePayment status transitions to DISPUTED. Escrowed funds for that milestone remain locked. Undisputed milestones continue normally. |
| 2 | Evidence submission window | Initiating party uploads evidence package: minimum 1 supporting document + written description of dispute. Photos/videos strongly recommended. Counter-party notified via email + portal and has 5 business days to submit counter-evidence. Either party can upload additional evidence within the window. | Both parties' evidence packages on record. 5-day response deadline set and tracked. |
| 3 | Platform Admin initial review | Admin Console shows Dispute Queue with all active disputes (priority-sorted by value and age). Admin reviews both evidence packages. Admin options: (a) Resolve directly for clear-cut cases, (b) Request additional evidence (extends timeline by 3 business days), (c) Escalate to Dispute Committee for complex cases. | Direct admin resolution OR escalation to Dispute Committee. |

### Phase B: Arbitration & Resolution

| Step | Action | System Behavior | Outcome |
|---|---|---|---|
| 4 | Dispute Committee convenes | Committee composition: 1 Raw Essentials representative + 1 independent trade finance expert + 1 industry vertical expert (Food/Energy/Vehicles as relevant). Committee reviews all evidence and may request additional information. 10 business day resolution SLA from committee formation. | Committee issues binding decision document with rationale. |
| 5a | Outcome: Full Release to Supplier | Dispute found unfounded. MilestonePayment.sol: DISPUTED > APPROVED > RELEASED. EscrowVault releases funds to supplier. Dispute fee (0.5% of disputed milestone value) charged to disputing buyer. Buyer credit score impact: -25 points for frivolous dispute. | Supplier paid in full. Buyer pays dispute fee. |
| 5b | Outcome: Partial Release | Partial quality/delivery issue confirmed. Committee determines fair split (e.g., 70% to supplier, 30% returned to pool). Both amounts released from escrow via separate transactions. Dispute fee split proportionally between parties. | Split payment executed on-chain. Both parties partially compensated. |
| 5c | Outcome: Full Refund to Buyer | Goods found materially deficient/non-conforming. EscrowVault.refund() returns funds to liquidity pool. Supplier responsible for goods retrieval at own cost. Dispute fee charged to supplier. Supplier credit score: -50 points. | Buyer/pool refunded. Supplier penalized. Trade NFT updated. |

| | | | |
|---|---|---|---|
| 5d | Outcome:<br>Re-inspection Required | Independent SGS Verax or equivalent inspection ordered. Cost initially split 50/50, reassigned by committee based on outcome. Funds remain locked during inspection (max 15 additional business days). Results feed back to committee for final decision. | Re-inspection results drive final resolution at Step 4. |

### DISPUTE FRAMEWORK RULES

WHO CAN DISPUTE: Buyer, Supplier, or Platform Admin (on compliance/fraud grounds).

WHEN: On milestones in PENDING_APPROVAL state, or within 5 business days after milestone release.

EVIDENCE REQUIRED: Minimum 1 document + written description. Photos/videos strongly recommended.

FUND HANDLING: Disputed milestone funds locked in EscrowVault. Undisputed milestones continue unaffected.

TIMELINE: Response 5 days + Admin review 5 days + Committee 10 days = Maximum 20 business days total.

ESCALATION: If Committee cannot resolve, external arbitration per UAE Federal Arbitration Law (No. 6 of 2018).

DISPUTE FEE: 0.5% of disputed milestone value, charged to losing party. Deters frivolous disputes.

CREDIT IMPACT: All outcomes affect both parties' credit scores. Pattern of disputes reduces future credit limits.

# 4. Feature Catalog

All features include Acceptance Criteria defining the Definition of Done. Every feature must satisfy its acceptance criteria before being marked complete in sprint review. Features are prioritized: P0 = Must Have for MVP launch, P1 = Important for pilot success.

## 4.1 Identity & Access Management (IAM)

| ID | Feature | Description | Pri | Acceptance Criteria |
|---|---|---|---|---|
| **IAM-01** | **User Registration** | Self-service registration for Providers and Traders. Entity type selection (Gold Provider, Cash Provider, Trader), email verification, profile completion with company details. | P0 | User registers, verifies email, selects entity type, completes profile. Duplicate email prevented. Registration event logged. Admin notified of new registration. |
| **IAM-02** | **KYC Integration** | Shufti Pro integration for identity verification, document verification (passport/Emirates ID), sanctions screening, PEP check. UAE entity verification: trade license validation, UBO declaration. | P0 | KYC flow completes end-to-end. Verified status written to DIDVerifier on-chain. Rejection reason displayed to user. Admin can view KYC status and override. Re-verification workflow works. |
| **IAM-03** | **Role-Based Access** | Five roles: Gold Provider, Cash Provider, Trader, Admin, Compliance Officer. Per-portal access control with documented permission matrix. | P0 | Each role can only access permitted portals/features. Unauthorized access returns 403. Role assignment requires admin approval. Permission matrix documented and enforced. |
| **IAM-04** | **MFA Authentication** | TOTP-based multi-factor authentication via authenticator apps. Mandatory for admin roles and all high-value operations exceeding $50K. | P0 | MFA enrollment works with Google Authenticator / Authy. High-value operations blocked without MFA. Recovery flow functional. Admin sessions always require MFA. |
| **IAM-05** | **On-Chain DID** | DIDVerifier.sol stores boolean isVerified flag per address. Maps platform user ID to blockchain wallet address. Only verified addresses can transact on-chain. | P0 | Verified users have on-chain flag true. Non-verified addresses rejected by all smart contracts. Mapping queryable. Revocation works for compliance actions. |
| **IAM-06** | **Wallet Provisioning** | Auto-generate HD wallet (BIP-44 path) on registration. Custodial model with HSM-managed keys (AWS CloudHSM). Unique address per user. No private key exposure. | P0 | Every registered user gets unique blockchain address. Keys stored in HSM. Derivation follows BIP-44. No private key visible in any log, API response, or database field. |

## 4.2 Asset Provider Operations (AP)

| ID | Feature | Description | Pri | Acceptance Criteria |
|---|---|---|---|---|
| **AP-01** | **Gold Deposit** | Multi-step workflow: gold details entry (weight, purity, custodian, vault), 4-document upload (Certificate of Ownership, Assay Report, Insurance, Custodian Confirmation), custodian verification (email-based MVP), compliance check, admin approval, on-chain registration, TCT minting. | P0 | Full flow from entry to TCT minting. All 4 documents uploadable (PDF, max 10MB each). Custodian verification email sent with template. Admin approval triggers on-chain registration. Rejection requires reason. |
| **AP-02** | **Cash Deposit** | Multi-step: amount entry (min $10K), source of funds document | P0 | Full flow from entry to CCT minting. Wire instructions displayed correctly. Banking API |

| | | | | |
|---|---|---|---|---|
| | | upload, wire instructions display (bank/account/reference), banking API receipt confirmation, AML screening, CCT token minting, LP token issuance. | | confirms receipt. AML screening runs before minting. LP tokens reflect correct pool share. |
| **AP-03** | **Gold Withdrawal** | Full or partial withdrawal request. Available liquidity check. FIFO queue if pool liquidity insufficient. Accumulated yield calculation. LP token + TCT burn. Settlement: fiat wire (T+1) or physical gold release (T+5 business days). | P0 | Withdrawal request created. Liquidity check accurate. FIFO queue activates correctly when pool insufficient. Token burn reflected on-chain. Wire/gold release initiated within SLA. |
| **AP-04** | **Cash Withdrawal** | Full or partial. Liquidity check. FIFO queue. Yield calculation to withdrawal date. LP + CCT burn. Fiat wire to registered bank account (T+1). | P0 | Same as AP-03 for cash. Wire initiated within T+1. Correct yield calculated. LP share reduced accurately. FIFO queue position displayed to user. |
| **AP-05** | **Provider Dashboard** | Real-time portfolio view: total deposited (gold oz / cash USD), current valuation (gold at LBMA price), LP share %, accumulated yield, projected APY, active trade exposure, withdrawal queue position (if any). | P1 | Dashboard loads within 3 seconds. All values accurate within 1-minute lag. Gold price updates every 15 minutes. Yield updates daily. Export to PDF functional. |

## 4.3 Trade Management (TM)

| ID | Feature | Description | Pri | Acceptance Criteria |
|---|---|---|---|---|
| **TM-01** | **Trade Application** | Upload trade documents (SBLC/ICPO/SCO/LOI, invoices, shipping docs), enter deal details (commodity, quantity, value, counterparty, proposed milestones). AI document verification. Application status tracking. | P0 | Application submits successfully. All document types uploadable. AI verification returns confidence score. Status visible in Trader Portal. Admin notified of new application. |
| **TM-02** | **Credit Scoring** | Risk assessment engine. MVP: Rules-based scoring with manual admin override (cold-start approach for 3 pilot deals with zero historical data). Factors: trade value, commodity type, counterparty profile, documentation quality. Future: XGBoost ML model. | P0 | Credit score generated for every application. Rules produce consistent results. Admin can override with documented reason. Score affects credit limit assignment. History logged for future ML training. |
| **TM-03** | **Milestone Payments** | Smart contract-managed 30/40/30 split. MilestonePayment.sol state machine: PENDING > APPROVED > RELEASED > COMPLETED (or DISPUTED). Escrow lock/release per milestone. Documentation verification at each gate. | P0 | 3 milestones created per trade. State transitions work correctly. Escrow locks/releases accurate amounts. Dispute transition functional. Cannot release without admin approval. |
| **TM-04** | **Trade NFT** | ERC-721 NFT minted per trade deal, encoding: deal value, parties, commodity, milestones, terms, status. Immutable trade record on-chain. Status updates as milestones progress. | P0 | NFT minted on deal creation. Metadata includes all deal terms. Status updates on milestone events. NFT queryable by all parties. Transfer restricted (soulbound to deal). |

| | | | | |
|---|---|---|---|---|
| **TM-05** | **Escrow Management** | EscrowVault.sol locks financing amount from pool. Per-milestone release authorization. Refund capability for disputes. Emergency admin release with multisig. | P0 | Funds lock correctly from pool. Release per milestone accurate to the wei. Refund works for dispute outcomes. Emergency release requires 3-of-5 multisig. |
| **TM-06** | **Repayment Tracking** | Track buyer repayment against financing terms. Automated reminders. Late payment detection and penalty calculation (2% per 30-day period). Credit score impact for late/missed payments. | P1 | Repayment schedule created on deal approval. Reminders sent at 7 days and 1 day before due. Late payment detected and penalty calculated accurately. Credit score updated on late event. |
| **TM-07** | **Chill Pay Off-Ramp** | Supplier token-to-fiat conversion. Integration model: platform redirects supplier to Chill Pay hosted checkout (not embedded). Supported currencies: AED, USD. Settlement timeline: T+1 for AED, T+2 for USD. Chill Pay handles their own KYC. | P1 | Supplier can initiate conversion from portal. Redirect to Chill Pay works. Conversion amount and currency correct. Settlement completes within SLA. Transaction status reflected back on platform. |

## 4.4 Compliance & Regulatory (CR)

| ID | Feature | Description | Pri | Acceptance Criteria |
|---|---|---|---|---|
| **CR-01** | **AML Screening** | Automated sanctions screening on every transaction. GOAML integration for UAE compliance reporting. Suspicious Transaction Report (STR) generation. Watchlist matching (UN, OFAC, EU). | P0 | Every deposit/withdrawal/trade triggers screening. Flagged transactions blocked pending review. GOAML-format reports exportable. STR generated with required fields. Watchlist updated weekly. |
| **CR-02** | **Transaction Monitoring** | Real-time monitoring for unusual patterns: velocity, amount thresholds, geographic risk, behavioral anomalies. Alert generation for compliance officer review. | P0 | Alerts generated for transactions exceeding thresholds. Compliance officer queue populated. False positive rate tracked. Alert resolution workflow complete. Audit trail for every alert. |
| **CR-03** | **Proof of Reserves** | Real-time dashboard comparing: on-chain token supply (TCT + CCT) versus off-chain verified reserves (gold + cash). Backing ratio displayed. Alert if ratio approaches 100%. | P0 | Dashboard accurate within 15-minute lag. Backing ratio calculation correct. Alert fires at 102% threshold. Historical backing ratio trend visible. Exportable for regulator reporting. |
| **CR-04** | **Risk Dashboard** | Unified risk view: pool utilization %, concentration per trade, default probability scores, provider exposure breakdown, active dispute count, regulatory compliance status. | P1 | Dashboard loads with all risk metrics. Pool utilization real-time. Concentration limit violations flagged. All data points sourced from live platform data. |
| **CR-05** | **Regulator Observer** | Read-only portal for VARA compliance auditors. Transaction browser, audit trail viewer, compliance report generator, real-time pool status. No write/modify capability. | P1 | Observer can log in with restricted role. All transactions viewable. Audit trail searchable by date/type/party. No edit/delete capabilities available. Session logging active. |
| **CR-06** | **Audit Trail** | Immutable log of every platform action: user actions, admin decisions, smart contract events, system events. Tamper-evident (hash-chained). Searchable and exportable. | P0 | Every action logged with timestamp, user, action type, before/after state. Logs hash-chained for tamper detection. Search by date range, user, action type. Export to CSV/PDF. |

## 4.5 Yield Engine (YE)

| ID | Feature | Description | Pri | Acceptance Criteria |
|---|---|---|---|---|
| **YE-01** | **Yield Calculation** | Daily calculation of provider yield from trade financing fees. Formula: Provider Yield = (LP Share % x 70% of Total Fees Collected) / Time Period. Compounding frequency: daily accrual, monthly distribution. | P0 | Yield calculated daily at UTC midnight. Accurate to 6 decimal places. Matches manual calculation within 0.01% tolerance. Handles partial-period deposits correctly. |
| **YE-02** | **Distribution Trigger** | Monthly automated yield distribution to all active LP holders. Pro-rata based on time-weighted average LP share. Distribution event logged on-chain. Provider notification sent. | P0 | Distribution executes monthly. All eligible providers receive correct amounts. On-chain distribution event emitted. Email/portal notification sent. Distribution history viewable. |
| **YE-03** | **LP Value Update** | Real-time LP token value calculation. Factors: underlying asset value (gold at LBMA + cash at par), accumulated yield, pool utilization. LP price displayed on Provider Dashboard. | P0 | LP value updates every 15 minutes. Reflects gold price changes, new deposits, withdrawals, and yield accrual. Historical LP value chart available. |

# 5. Blockchain Architecture

## 5.1 Why Avalanche Evergreen Subnet

After evaluating Polygon Supernets, OP Stack, Arbitrum Orbit, and Hyperledger Besu, Avalanche Evergreen Subnet was selected for the following reasons: native EVM compatibility (Solidity smart contracts, standard tooling), sub-2-second finality (critical for trade finance), permissioned validator set (VARA KYC compliance), configurable gas economics (can set gas to zero for users), enterprise-grade support through Ava Labs, and proven institutional adoption track record.

| Decision | Detail |
|---|---|
| **Network Type** | Private Permissioned Avalanche Evergreen Subnet - closed-loop, all validators are known and KYC-verified entities |
| **Consensus** | Snowman Consensus (Avalanche family) - optimized for linear chain ordering, sub-2-second finality |
| **EVM Version** | Subnet-EVM - full Ethereum Virtual Machine compatibility. All Solidity tooling works: Hardhat, Foundry, OpenZeppelin |
| **Validators** | Minimum 4 validator nodes in controlled environment. Raw Essentials operates 2, partner entities operate 2. All validators KYC-verified. |
| **Gas Model** | Zero-gas for end users. Platform covers gas costs from operational budget. Users never need native tokens for transactions. |
| **Block Size** | 15M gas limit per block. Sufficient for ~500 transactions per block at average complexity. |
| **Hosting** | AWS me-south-1 (Bahrain). Validator nodes on EC2 c5.2xlarge instances. Data residency compliant. |

## 5.2 Smart Contract Architecture

All contracts use battle-tested OpenZeppelin standards following the 'Simple & Safe' principle: complexity is the enemy of security. In a controlled private blockchain environment, proven approaches provide better security through simplicity than cutting-edge but unproven technologies.

| Contract | Standard | Purpose |
|---|---|---|
| **AssetTokenFactory** | ERC-20 + AccessControl + Pausable + UUPSUpgradeable | Manages TCT (gold) and CCT (cash) token minting/burning. registerGold() and registerCash() functions. Only callable by verified addresses with MINTER_ROLE. |
| **LiquidityPool** | ERC-20 (LP Token) + ReentrancyGuard + UUPSUpgradeable | Unified liquidity pool. deposit() accepts TCT or CCT, issues LP tokens. withdraw() burns LP, returns underlying. Tracks pool composition and backing ratio. |
| **EscrowVault** | AccessControl + ReentrancyGuard + Pausable | Locks trade financing amounts from pool. Per-trade escrow tracking. release() for milestone completion. refund() for disputes. Emergency multisig release. |
| **MilestonePayment** | AccessControl + Pausable | State machine per milestone: PENDING > APPROVED > RELEASED (or DISPUTED > resolution). Manages 30/40/30 split logic. Emits events for all transitions. |
| **TradeNFT** | ERC-721 + AccessControl | Non-transferable (soulbound) NFT per trade deal. Stores deal metadata hash. Status field tracks trade lifecycle. Queryable by all parties. |
| **InvoiceNFT** | ERC-721 + AccessControl | Represents verified trade documents. AI verification results stored as metadata. Linked to parent TradeNFT. |

| | | |
|---|---|---|
| **DIDVerifier** | AccessControl | Maps platform user IDs to blockchain addresses. Stores boolean isVerified flag. All other contracts check this before allowing transactions. |
| **CreditScore** | AccessControl + UUPSUpgradeable | On-chain credit score storage. Updated by backend after off-chain calculation. getScore() used by LiquidityPool for credit limit enforcement. |
| **YieldDistributor** | AccessControl + ReentrancyGuard | Calculates and distributes yield to LP holders. Monthly distribution trigger. Pro-rata based on LP share percentage. |

## 5.3 On-Chain vs Off-Chain Boundary

Clear separation: on-chain for trust-critical operations (tokenization, escrow, settlement, audit trail), off-chain for everything else (user data, KYC documents, business logic, AI processing, reporting). This boundary is immutable for the project - changes require formal Change Request.

| ON-CHAIN (Blockchain) | OFF-CHAIN (Backend/Cloud) |
|---|---|
| Token minting and burning (TCT, CCT, LP) | User registration, profiles, KYC document storage |
| Liquidity pool deposits and withdrawals | KYC verification (Shufti Pro API calls) |
| Escrow lock, release, and refund | AI document verification processing |
| Milestone state transitions | Credit scoring calculation (rules engine / ML) |
| Trade NFT minting and status updates | Banking API integrations (wire confirmation) |
| DID verification status (boolean flags) | Gold custodian email verification workflow |
| Credit score storage (numeric value only) | Chill Pay off-ramp redirect and settlement tracking |
| Yield distribution events | Notification system (email, push) |
| Proof of Reserves backing ratio | Reporting, dashboards, analytics |
| All event emissions (audit trail) | GOAML report generation and submission |

# 6. Token Architecture & Oracle Strategy

## 6.1 Token Types

| Token | Standard | Represents | Minting / Burning Rules |
|---|---|---|---|
| **TCT (Trade Collateral Token - Gold)** | ERC-20 | 1 TCT = 1 troy ounce gold equivalent at LBMA spot price. Represents tokenized physical gold held by verified custodians. | MINT: On verified gold deposit (admin-approved, custodian-confirmed). BURN: On gold withdrawal (after liquidity check and yield settlement). |
| **CCT (Cash Collateral Token)** | ERC-20 | 1 CCT = 1 USD. Represents verified cash deposits confirmed via banking API. Stable value. | MINT: On wire receipt confirmation from banking API + AML clearance. BURN: On cash withdrawal (T+1 wire settlement). |
| **LP Token** | ERC-20 | Represents proportional share of the unified liquidity pool. Value = (Total Pool Value in USD) / (Total LP Supply). Basis for yield distribution. | MINT: On any deposit (gold or cash). Amount = deposit USD value / current LP price. BURN: On withdrawal. Redeems proportional pool share. |
| **TradeNFT** | ERC-721 | Unique trade deal record. Non-transferable (soulbound). Metadata: deal value, parties, commodity, milestones, status, terms. | MINT: On trade deal approval and funding. One per trade. Never burned - permanent record. Status field updated through lifecycle. |
| **InvoiceNFT** | ERC-721 | Verified trade document record. Linked to parent TradeNFT. Metadata: document hash, AI verification score, document type. | MINT: On document verification completion. Multiple per trade (one per verified document). Never burned. |

## 6.2 Gold Price Oracle Strategy

Gold pricing is critical - it determines TCT valuation, pool value, LP prices, and backing ratios. The oracle strategy prioritizes reliability and tamper resistance appropriate for a $16M pilot.

| Component | Detail |
|---|---|
| **Primary Source** | LBMA Gold Price (London Bullion Market Association) - the global benchmark for gold pricing used by central banks and institutional investors. AM and PM fix. |
| **Feed Method** | Backend service fetches LBMA price via licensed data provider API every 15 minutes. Admin can trigger manual update. Price written to on-chain PriceOracle contract by authorized ORACLE_ROLE address. |
| **Circuit Breaker** | If new price deviates >5% from previous on-chain price, update is held pending admin review. Prevents flash crash or manipulation from affecting pool valuations. Admin can approve or reject the update. |
| **Staleness Protection** | If on-chain price is older than 30 minutes, all gold-dependent operations (new deposits, new trades) are paused until fresh price is available. Withdrawals continue at last known price. |
| **Fallback Sources** | If primary feed fails: (1) Secondary API provider, (2) Manual admin entry with dual-approval (two admins must confirm), (3) Operations pause if no price available for >2 hours. |
| **Audit Trail** | Every price update logged on-chain: timestamp, source, old price, new price, updater address. Immutable record for VARA compliance and Proof of Reserves verification. |

# 7. Liquidity Pool Mechanics

This section addresses a critical architectural question: how do gold-backed (TCT) and cash-backed (CCT) tokens coexist in a single pool, and how are trade deals financed from this pool? This is the core business mechanism of the entire platform.

## 7.1 Pool Structure: Unified Pool with Dual-Asset Composition

**DESIGN DECISION**

ONE UNIFIED POOL - not two separate pools. TCT and CCT tokens are deposited into a single LiquidityPool contract.

Pool value is calculated in USD equivalent: CCT at par ($1 each) + TCT at LBMA gold price.

LP tokens represent proportional share of the TOTAL pool regardless of which asset type was deposited.

This means a Gold Provider's LP share gives them exposure to the entire pool's performance, not just gold.

Rationale: Unified pool maximizes liquidity, simplifies trade financing, and distributes risk across all providers.

## 7.2 Pool Valuation

Total Pool Value (USD) = (TCT Supply x LBMA Gold Price per oz) + (CCT Supply x $1.00). Example: If pool contains 500 TCT (gold at $2,000/oz) and 1,000,000 CCT, Total Pool Value = (500 x $2,000) + (1,000,000 x $1) = $1,000,000 + $1,000,000 = $2,000,000.

LP Token Price = Total Pool Value / Total LP Supply. If 2,000,000 LP tokens outstanding, LP Price = $1.00 each. When gold price rises to $2,200/oz: New Pool Value = (500 x $2,200) + $1,000,000 = $2,100,000. LP Price rises to $1.05. All LP holders benefit from gold appreciation.

## 7.3 How Trades Are Financed

| Step | Action | Details |
|---|---|---|
| STEP 1 | System checks available liquidity | Total Pool Value - Already Locked in Escrow; 25% concentration limit (no single trade > 25% of pool) |
| STEP 2 | Required CCT locked from pool into EscrowVault | If pool has sufficient CCT balance, direct lock; If CCT insufficient but total pool value sufficient (due to TCT), system uses CCT first and flags remaining as requiring gold-backed collateral (TCT not directly disbursed, stays as backing) |
| STEP 3 | Milestone releases send CCT from escrow to supplier's wallet | Supplier converts CCT to fiat via Chill Pay |
| STEP 4 | Buyer repayment returns CCT to pool | Restores liquidity; Financing fee collected and fed to YieldDistributor |

## 7.4 Pool Utilization and Capacity

| Metric | Definition & Threshold |
|---|---|
| **Total Pool Capacity** | Sum of all TCT (at LBMA price) + CCT in the pool. Represents maximum theoretical financing available. |
| **Available Liquidity** | Total Pool Capacity minus all funds currently locked in EscrowVault across active trades. This is what's available for new trades. |

| | |
|---|---|
| **Utilization Rate** | Locked in Escrow / Total Pool Capacity. Target: 40-70%. Below 40% = underutilized (providers earning less yield). Above 80% = liquidity risk alert. |
| **Concentration Limit** | Maximum 25% of Total Pool Capacity per single trade. Prevents catastrophic loss from single default. Hard-enforced in LiquidityPool contract. |
| **Reserve Ratio** | Minimum 15% of Total Pool must remain unlocked at all times to service withdrawal requests. If utilization approaches 85%, new trade approvals paused. |

# 8. Fee Structure & Revenue Model

## 8.1 Fee Breakdown

| Fee Type | Rate | Description |
|---|---|---|
| **Trade Financing Fee** | 3-5% of trade value | Primary revenue source. Charged to buyer upon deal approval. Deducted from financing amount before escrow lock. Variable based on credit score and trade risk profile. |
| **Late Payment Penalty** | 2% per 30-day period | Applied to overdue repayments. Accrues monthly. Funds default recovery waterfall if payment not received within grace period. |
| **Dispute Fee** | 0.5% of disputed milestone | Charged to losing party in dispute resolution. Deters frivolous disputes while keeping cost reasonable for legitimate claims. |
| **Early Withdrawal Fee** | 0.25% if < 30 days | Applies to provider withdrawals within 30 days of deposit. Discourages short-term parking. Waived after 30 days. |

## 8.2 Fee Distribution

Every trade financing fee collected is distributed as follows:

| Allocation | Percentage | Purpose |
|---|---|---|
| **Provider Yield** | 70% | Distributed to LP holders proportional to their pool share. This is the primary incentive for providers to deposit assets. Target: 6-8% APY. |
| **Platform Revenue** | 20% | Raw Essentials operating revenue. Covers team costs, infrastructure, compliance, business development. |
| **Reserve Fund** | 5% | First-loss protection fund. Used to cover provider losses in case of trader default (see Section 9 Default Handling). Target: accumulate 5% of total pool value. |
| **Operations** | 5% | Infrastructure costs (AWS, blockchain nodes, HSM), third-party services (KYC, oracle feeds), insurance premiums. |

# 9. Default & Non-Payment Handling

This section defines what happens when a trader fails to repay financing - the single highest-risk scenario for the platform and its providers. In a $16M pilot, even one default on the $8M Energy deal would be catastrophic without a clear recovery framework. Every stakeholder (providers, traders, admins, legal) must understand the exact process before a single dollar enters the pool.

## 9.1 Default Timeline

| Phase | Timeline | Actions |
|---|---|---|
| **Grace Period** | Repayment due date + 7 calendar days | Automated reminders at due date, +3 days, +5 days, +7 days. No penalties during grace period. Buyer can still make payment through normal channels. Admin notified at +3 days. |
| **Late Payment** | Day 8 through Day 60 | 2% late payment penalty applied (per 30-day period). Credit score reduced: -25 points at Day 8, additional -25 at Day 30. Future credit limit reduced by 50%. Admin escalation at Day 30. Weekly outreach to buyer. |
| **Pre-Default Notice** | Day 61 - Formal notice issued | Formal pre-default notice sent to buyer via email and registered mail. 15 calendar day cure period. Buyer's platform access restricted to repayment function only. All future trade applications frozen. |
| **Default Declaration** | Day 76 - Official default | Trade status: DEFAULTED. Default recovery waterfall activated (Section 9.2). Credit score: 0 (blacklisted). Platform access revoked. VARA regulatory reporting triggered. Legal proceedings initiated. |

## 9.2 Default Recovery Waterfall

When a default occurs, recovery follows a strict priority order. Each level is exhausted before moving to the next. This waterfall protects providers by ensuring multiple layers of protection absorb losses before provider capital is affected.

| Priority | Source | Coverage | Details |
|---|---|---|---|
| **1** | Late Payment Penalties | Variable | Accumulated late fees from the defaulting trade. Typically 2-6% of outstanding amount depending on how long the default timeline ran. |
| **2** | Reserve Fund | Up to 5% of pool value | Built from 5% allocation of all financing fees. Continuously replenished from ongoing fee income. Covers first-loss position protecting providers. |
| **3** | Trade Credit Insurance | Per policy terms | Insurance claim against trade credit insurance policy (if obtained). Coverage typically 70-90% of outstanding. Procurement of insurance is a pre-launch activity for Gary. |
| **4** | Pro-Rata Provider Absorption | Remaining shortfall | After exhausting priorities 1-3, remaining loss distributed pro-rata across all LP holders based on their pool share percentage. Transparent: loss amount shown on Provider Dashboard. |
| **5** | Legal Recovery | Ongoing | Legal proceedings against defaulting trader to recover outstanding amount. UAE commercial courts. Any recovered funds distributed to providers who absorbed losses. Recovery timeline: 6-18 months typical. |

## 9.3 Provider Protection Mechanisms

Multiple layers prevent catastrophic provider loss: Milestone-based release (30/40/30) limits maximum exposure at any point in a trade - the worst case is 30% of a deal value, not 100%. Concentration limit (25% max per trade) prevents a single default from devastating the entire pool. Credit scoring with tiered limits caps exposure per trader based on risk profile. The Reserve Fund provides first-loss protection built from platform fees. The Provider Portal displays real-time risk metrics including default probability per active trade.

## 9.4 Regulatory Reporting

All defaults are reportable events under VARA compliance framework. Platform must report: default amount and outstanding balance, recovery actions taken and timeline, provider impact (loss absorption amounts), resolution timeline and legal proceedings status. GOAML (UAE FIU) reporting triggered automatically if default involves any suspicious activity indicators (unusual patterns, sanctions nexus, geographic risk flags).

# 10. UI Portal Definitions

The platform consists of four distinct portals, each tailored to its user persona. All portals share common design principles: responsive web design (desktop-first, mobile-adaptive), consistent design system with shared component library, real-time data via WebSocket connections, session management with automatic timeout (30 minutes idle, 8 hours absolute), and accessibility compliance (WCAG 2.1 AA).

## 10.1 Provider Portal

Primary interface for Gold Providers and Cash Providers to manage their assets, monitor yield, and track trade exposure.

| Screen | Key Functionality |
|---|---|
| **Dashboard** | Portfolio overview: total deposits, current valuation, LP share %, accumulated yield, projected APY, active trade exposure, recent transactions. Gold price ticker with 15-min updates. |
| **Deposit Gold** | Multi-step wizard: gold details entry (weight, purity 995/999, custodian, vault), 4-document upload with drag-drop (CoO, Assay, Insurance, Custodian Confirmation), review & submit. Progress stepper visible throughout. |
| **Deposit Cash** | Amount entry (min $10K), source of funds upload, wire instructions display with copy-to-clipboard, deposit status tracker showing banking API confirmation. |
| **Withdrawals** | Full or partial withdrawal request, available balance display, FIFO queue position (if applicable), accumulated yield breakdown, settlement method selection (fiat wire / physical gold). |
| **Yield Report** | Daily yield accrual history, yield source breakdown (which trades generated yield), projected future yield, historical APY chart, export to CSV/PDF. |
| **Trade Exposure** | Active trades funded from pool, exposure per trade, milestone progress, concentration metrics, risk indicators. Read-only view (providers cannot intervene in trades). |

## 10.2 Trader Portal

Interface for Traders (Buyers and Suppliers) to create deals, manage trades, track milestones, and handle invoicing.

| Screen | Key Functionality |
|---|---|
| **Dashboard** | Active trades summary, pending actions (approvals, document uploads), upcoming milestones, recent notifications, credit score indicator (rules-based for MVP). |
| **Create Deal** | Deal formation wizard: counterparty selection (search verified traders), commodity details, quantity, pricing, delivery terms (Ex Works / CIF / FOB), payment terms, required documents checklist, milestone definition. |
| **Trade Lifecycle** | Visual pipeline: Deal Created > Escrow Funded > Goods Shipped > Goods Inspected > Milestone Released > Trade Complete. Each stage shows required actions, documents, and approval status. |
| **Invoice Management** | Create invoice against trade, attach supporting documents, submit for approval, track payment status. InvoiceNFT minted on submission. Invoice history with filtering and search. |
| **Dispute Center** | Raise dispute with category selection, evidence upload, timeline tracker through 2-phase process (Initiation > Arbitration), resolution history, dispute fee acknowledgment. |
| **Documents** | Trade document repository: upload, version history, verification status, document type tagging (SBLC, ICPO, SCO, LOI, Bill of Lading, Phytosanitary, SGS Verax, FSSAI). Secure sharing with counterparty. |

## 10.3 Admin Console

Comprehensive management interface for platform administrators and compliance officers. All admin actions are audit-logged with timestamp, actor, action, and target.

| Screen | Key Functionality |
|---|---|
| **Operations Dashboard** | Platform health metrics: active users, pending KYC, active trades, pool utilization, default rates, revenue summary. Real-time alerts for threshold breaches (pool utilization >80%, high-value transactions >$500K). |
| **User Management** | User list with status filters, KYC review queue, role assignment, account suspension/reactivation, compliance action log. Bulk operations for audit efficiency. |
| **KYC Review** | Queue of pending verifications, document viewer (side-by-side comparison), Shufti Pro results display, approve/reject with mandatory reason, escalation to compliance officer for edge cases. |
| **Trade Oversight** | All active trades with status, intervention controls (pause trade, force milestone, initiate dispute), trade value and risk metrics, exception management queue. |
| **Pool Management** | Liquidity pool dashboard: total value, asset split (TCT/CCT), utilization rate, reserve ratio, concentration metrics. Reserve fund balance. Withdrawal queue management. |
| **Compliance Center** | AML alert queue, STR generation and GOAML submission, sanctions screening results, VARA reporting dashboard, audit trail search and export, regulatory filing calendar. |
| **System Config** | Fee configuration, threshold management, oracle parameters, circuit breaker controls, smart contract admin functions (pause, unpause), validator node health monitoring. |

## 10.4 Regulator Observer Portal

Read-only portal for VARA regulators and authorized auditors providing transparency into platform operations without any ability to modify data or trigger actions.

| Screen | Key Functionality |
|---|---|

| | |
|---|---|
| **Platform Overview** | Aggregate statistics: total platform TVL, transaction volume, user count by type, trade count by status. No individual user data visible without explicit query. |
| **Transaction Explorer** | Searchable transaction log with filters (date range, type, value range, status). On-chain transaction hash linked to Avalanche subnet explorer. Export to regulatory reporting format. |
| **Compliance Reports** | Pre-built regulatory reports: monthly activity summary, AML screening results, KYC statistics, default and recovery metrics. Scheduled auto-generation and delivery. |
| **Audit Trail** | Immutable log of all platform actions. Filter by user, action type, time range. Blockchain-anchored hash for tamper evidence. Export for external audit tools. |

# 11. Non-Functional Requirements

Non-functional requirements define quality attributes the platform must exhibit. Each NFR has measurable criteria and verification methods.

## 11.1 Performance

| Requirement | Target | Measurement |
|---|---|---|
| API response time (P95) | < 500ms for standard operations | Application Performance Monitoring (APM) dashboards, load testing with k6/JMeter |
| Dashboard page load | < 3 seconds (first meaningful paint) | Lighthouse CI, real user monitoring (RUM) |
| Blockchain transaction confirmation | < 2 seconds (sub-second target) | On-chain monitoring, transaction receipt timing |
| Concurrent users | 100 simultaneous users for MVP pilot | Load testing with simulated user sessions |
| Oracle price update | Every 15 minutes from LBMA feed | Oracle monitoring, staleness alerts at 30 mins |
| Report generation | < 30 seconds for standard reports | Background job monitoring, user notification on completion |

## 11.2 Security

| Requirement | Specification |
|---|---|
| **Encryption at rest** | AES-256 for all stored data. AWS KMS managed keys. Database-level encryption (RDS TDE). File storage encryption (S3 SSE-KMS). |
| **Encryption in transit** | TLS 1.3 for all API communications. Certificate pinning for mobile clients. Mutual TLS (mTLS) for service-to-service communication within the platform. |
| **Key management** | AWS CloudHSM (FIPS 140-2 Level 3) for blockchain private keys. Key rotation policy: 90 days for application keys, annual for HSM master keys. No private key material in logs or backups. |
| **Authentication** | JWT with RS256 signing. Access token: 15-minute expiry. Refresh token: 7-day expiry with rotation. TOTP-based MFA mandatory for admin and high-value operations (>$50K). |
| **Authorization** | Role-based access control (RBAC) with 5 roles. On-chain AccessControl.sol for smart contract functions. API gateway enforces role permissions before request routing. |

| | |
|---|---|
| **Smart contract security** | OpenZeppelin battle-tested contracts. ReentrancyGuard on all state-changing functions. Pausable emergency stop. UUPS upgradeable proxy pattern. Third-party audit before mainnet deployment. |
| **Penetration testing** | Pre-launch penetration test by certified third party. Quarterly vulnerability assessments post-launch. Automated SAST/DAST in CI/CD pipeline. |

## 11.3 Availability & Disaster Recovery

| Requirement | Specification |
|---|---|
| **Uptime SLA** | 99.5% availability (excluding planned maintenance windows). Planned maintenance: maximum 4 hours per month, scheduled outside UAE business hours (weekends/nights). |
| **Recovery Point Objective (RPO)** | < 1 hour for off-chain data. Zero for on-chain data (blockchain is the source of truth; replicated across all validator nodes). |
| **Recovery Time Objective (RTO)** | < 4 hours for full platform recovery. < 1 hour for individual service recovery. Hot standby for critical services (API gateway, blockchain nodes). |
| **Backup strategy** | Database: automated daily snapshots retained 30 days. Transaction logs: continuous backup with 5-minute intervals. Document storage: cross-region replication (primary UAE, secondary Singapore). |
| **Blockchain resilience** | Minimum 4 validator nodes across 2 availability zones. Consensus tolerates 1 node failure (3 of 4 for >67% BFT threshold). Node recovery automated via infrastructure-as-code. |

## 11.4 Scalability

MVP is designed for pilot scale with architecture that supports horizontal scaling for production growth.

| Dimension | MVP Pilot Target | Production Ready Architecture |
|---|---|---|
| Users | 50-100 active users | Auto-scaling application tier, database read replicas |
| Trades | 3 concurrent trades ($16M total) | Event-driven architecture, async processing queues |
| Transactions | 1,000 blockchain TXs per day | Avalanche subnet supports 4,500+ TPS |
| Storage | 100GB documents, 10GB database | S3 unlimited storage, database auto-scaling |

## 11.5 Compliance

| Requirement | Specification |
|---|---|
| **Data residency** | All PII and financial data stored within UAE (AWS Middle East - Bahrain region or UAE local data center). No cross-border data transfer without explicit regulatory approval. |
| **Audit trail** | Immutable audit log for all user actions, system events, and administrative operations. Minimum 7-year retention per UAE financial regulations. Blockchain-anchored hash for tamper evidence. |
| **Data protection** | Compliance with UAE Federal Decree-Law No. 45 of 2021 on Personal Data Protection. PII encryption, access controls, data minimization, right to erasure (with regulatory exemptions for financial records). |

| | |
|---|---|
| **Regulatory reporting** | Automated GOAML STR filing. VARA periodic reporting (monthly transaction summaries, quarterly compliance reports). Central Bank UAE reporting for significant transactions. |

# 12. Operational Scope

This section defines the operational boundaries of the MVP deployment, including infrastructure, monitoring, and support.

## 12.1 Deployment Architecture

MVP deployment uses a cloud-native architecture on AWS with the Avalanche Evergreen Subnet as the blockchain layer.

| Component | Deployment Specification |
|---|---|
| **Cloud Provider** | AWS (Middle East - Bahrain region for data residency compliance). Multi-AZ deployment for high availability. |
| **Containerization** | Docker containers orchestrated via AWS ECS Fargate (serverless). No EC2 instance management overhead. Auto-scaling based on CPU/memory utilization. |
| **Blockchain Nodes** | 4 Avalanche validator nodes across 2 AZs. Dedicated EC2 instances (c5.xlarge minimum) with SSD storage. Managed by platform team. |
| **Database** | AWS RDS PostgreSQL with Multi-AZ. Encrypted storage (AES-256). Automated backups with 30-day retention. Read replica for reporting queries. |
| **Document Storage** | AWS S3 with server-side encryption (SSE-KMS). Versioning enabled. Lifecycle policies for archival (90 days to Glacier). Cross-region replication to Singapore. |
| **CDN / API Gateway** | AWS CloudFront for static assets. AWS API Gateway for REST endpoints with rate limiting, request validation, and WAF integration. |
| **Key Management** | AWS CloudHSM for blockchain private keys (FIPS 140-2 Level 3). AWS KMS for application encryption keys. Secrets Manager for API credentials. |

## 12.2 Monitoring & Alerting

Comprehensive monitoring across all platform layers with defined escalation procedures.

| Monitor | Tool | Alert Threshold |
|---|---|---|
| Application Performance | AWS CloudWatch + X-Ray distributed tracing | API P95 > 1s, error rate > 1%, 5xx spike |
| Blockchain Health | Custom node health checks + CloudWatch | Node down, block gap > 5s, consensus failure |
| Oracle Freshness | Scheduled Lambda health check every 5 minutes | Price stale > 30 min, deviation > 5% from backup |
| Security Events | AWS GuardDuty + CloudTrail + WAF logs | Failed login > 5 attempts, unusual API patterns, WAF blocks |
| Pool Utilization | Custom dashboard with real-time metrics | Utilization > 80%, reserve < 15%, concentration > 25% |
| Infrastructure | AWS CloudWatch infrastructure metrics | CPU > 80%, memory > 85%, disk > 75%, unhealthy targets |

## 12.3 Support & Maintenance

**Support Hours:** UAE business hours (Sunday-Thursday, 8 AM - 6 PM GST) for pilot phase.

**Incident Response:** P1 Critical (platform down): 30-minute response, 4-hour resolution target. P2 High (feature degraded): 2-hour response, 24-hour resolution. P3 Medium: next business day. P4 Low: scheduled maintenance window.

**Maintenance Windows:** Saturday evenings (8 PM - 12 AM GST). Maximum 4 hours per month. Emergency patches may be deployed outside window with stakeholder notification.

**Change Management:** All production changes require: change request with rollback plan, peer review for code changes, staging environment validation, approval from Technical Lead (standard) or CTO (emergency).

## 12.4 Quality Assurance Scope (F-14)

QA strategy ensures comprehensive coverage across all platform layers with clearly defined responsibilities, test types, and entry/exit criteria.

### 12.4.1 Testing Strategy

| Test Type | Scope | Responsibility | Tooling |
|---|---|---|---|
| Unit Tests | Individual functions and methods | Development team (per-module) | Jest (backend), Hardhat (smart contracts), React Testing Library (frontend) |
| Integration Tests | API endpoints, DB queries, service interactions | Development team + QA | Supertest, Postman collections, Docker Compose test environments |
| Smart Contract Tests | All contract functions, edge cases, access control | Blockchain developer + external auditor | Hardhat, Slither (static analysis), Foundry fuzz testing |
| E2E Tests | Critical user journeys (deposit, trade, withdrawal) | QA team | Cypress or Playwright, automated regression suite |
| Performance Tests | Load, stress, endurance testing | QA + DevOps | k6 or JMeter, AWS load testing infrastructure |
| Security Tests | OWASP Top 10, smart contract vulnerabilities | External security firm + internal review | OWASP ZAP (DAST), Snyk (SCA), manual penetration test |
| UAT | Business acceptance by stakeholders | Product owner + selected pilot users | Structured test scripts, feedback forms, recorded sessions |

### 12.4.2 Coverage Requirements

**Unit Test Coverage:** Minimum 80% line coverage for backend services, 90% for smart contracts. Coverage reports generated in CI pipeline and visible in merge request reviews.

**Smart Contract Audit:** External audit by certified blockchain security firm required before mainnet deployment. All critical and high findings must be resolved; medium findings require documented mitigation plan.

**E2E Coverage:** All 4 primary user journeys (Section 3.2) must have automated E2E tests. Regression suite must pass before any production deployment.

**UAT Approach:** 2-week dedicated UAT phase with structured test scenarios mapped to acceptance criteria. Minimum 3 pilot users per persona (Provider, Trader). Exit criteria: zero P1 defects, fewer than 5 P2 defects with documented workarounds.

### 12.4.3 Environment Strategy

| Environment | Purpose & Configuration |
|---|---|

| | |
|---|---|
| **Development** | Individual developer environments with local blockchain (Hardhat node). Auto-deployed on merge to feature branches. Synthetic test data. |
| **Staging** | Mirror of production with Avalanche testnet subnet. Integration testing with mock banking APIs and test oracle. Production-equivalent infrastructure at reduced scale. |
| **UAT** | Dedicated environment for stakeholder acceptance testing. Production-equivalent data with anonymized real-world scenarios. Accessible to pilot users with test credentials. |
| **Production** | Live environment with Avalanche Evergreen Subnet. Full security controls, monitoring, and alerting. Deployment via CI/CD with manual approval gate. |

# 13. External Integrations

This section defines all external system integrations required for MVP Phase 0. Each integration includes scope, status, and the approach for the current phase. The platform follows a manual-first integration philosophy for MVP: manual processes with email-based workflows are used where API integration is not yet confirmed or adds unnecessary risk for pilot scale.

## 13.1 Integration Status Overview (F-12)

| Integration | Partner | Status | MVP Approach | Future Phase |
|---|---|---|---|---|
| **Payment Gateway** | Chill Pay | Confirmed | API Integration | Full automation, multi-currency |
| **KYC/AML Provider** | Shufti Pro | Confirmed | API Integration | Enhanced screening, ongoing monitoring |
| **Gold Price Oracle** | LBMA / DataScope | Confirmed | API Integration | Multi-source aggregation, real-time feeds |
| **Gold Custodian** | TBD (Dubai-based) | In Progress | Email-based verification | API integration for real-time verification |
| **Banking Partner** | TBD (UAE licensed bank) | In Progress | Manual wire confirmation | Open Finance API, real-time settlement |
| **Insurance Provider** | TBD | Not Started | Manual policy upload | API for policy verification and claims |
| **AML Reporting** | UAE FIU (GOAML) | Confirmed | Manual STR filing | Automated GOAML XML submission |
| **Blockchain Infra** | Ava Labs / AvaCloud | Confirmed | Evergreen Subnet | Multi-subnet architecture, cross-chain bridges |

## 13.2 Chill Pay Integration (F-09)

Chill Pay is the confirmed payment gateway partner for fiat on/off ramp operations. This integration is critical for cash deposit confirmations and withdrawal disbursements.

### 13.2.1 MVP Scope

**Wire Receipt Confirmation:** API webhook notification when incoming wire transfer matches expected reference number and amount. Platform uses this confirmation to trigger CCT minting.

**Withdrawal Disbursement:** API call to initiate outgoing wire transfer to provider's registered bank account. Supports T+1 settlement for standard withdrawals.

**Transaction Reconciliation:** Daily reconciliation report (API or SFTP) matching platform records against Chill Pay's ledger. Automated discrepancy detection with admin alerts.

**Supported Currencies:** USD (primary), AED (secondary). Multi-currency support deferred to future phase.

**Security:** Webhook signature verification (HMAC-SHA256), IP whitelisting, API key rotation (90-day cycle), encrypted payload for sensitive fields.

### 13.2.2 Integration Architecture

**Protocol:** REST API over HTTPS (TLS 1.3). Webhook callbacks for asynchronous events.

**Authentication:** API key + secret with HMAC request signing. Separate credentials for sandbox and production.

**Error Handling:** Exponential backoff retry (3 attempts, 1s/5s/30s intervals). Dead letter queue for failed webhooks. Admin notification on persistent failures. Manual reconciliation fallback.

**Sandbox Testing:** Full sandbox environment available for integration testing. Simulated wire transfers with configurable delays and failure scenarios.

## 13.3 Shufti Pro KYC Integration

**Scope:** Identity verification (passport, Emirates ID), document verification, sanctions screening (OFAC, UN, EU, UAE local lists), PEP check, AML risk scoring.

**Flow:** User initiates KYC from platform > redirected to Shufti Pro hosted verification > Shufti Pro processes and returns result via webhook > platform updates user status and writes DID on-chain.

**UAE-Specific:** Trade license verification for entities, UBO (Ultimate Beneficial Owner) declaration and verification, Emirates ID OCR and validation.

## 13.4 Gold Price Oracle

**Primary Source:** LBMA Gold Price (London Bullion Market Association) - the global benchmark for gold pricing. Fetched via licensed data feed (DataScope/Refinitiv API).

**Update Frequency:** Every 15 minutes during trading hours. Twice-daily LBMA fix (10:30 AM and 3:00 PM London time) used as anchor price.

**Circuit Breaker:** 5% deviation from last confirmed price triggers automatic pause on all gold-related operations (minting, trading, withdrawals). Admin review required to resume.

**Fallback Strategy:** Primary: LBMA feed. Secondary: Kitco/Bloomberg API. Tertiary: Manual admin price entry with 4-eye approval (two admins must confirm). Staleness protection: prices older than 30 minutes are rejected for any token minting or settlement.

# 14. Regulatory Compliance

Raw Essentials operates within Dubai's regulatory framework as defined by the Virtual Assets Regulatory Authority (VARA) and UAE federal financial regulations. Compliance is not an afterthought but a core design principle embedded in every layer of the platform.

## 14.1 VARA Licensing

VARA, established under Dubai Law No. 4 of 2022, is the world's first independent virtual asset regulator. The platform must obtain a VARA license before commercial operations. VARA Version 2.0 Rulebooks, effective June 19, 2025, introduce stricter compliance requirements.

### 14.1.1 Licensing Pathway

| Step | Phase | Description |
|---|---|---|
| 1 | Initial Disclosure (IDQ) | Submit Initial Disclosure Questionnaire: business plan, technology architecture, governance structure, AML/CFT framework, financial projections. 30-60 day review. |
| 2 | Initial Approval | VARA grants conditional approval with specific requirements. Company must establish UAE entity, appoint compliance officer, implement required controls. |
| 3 | Preparatory Phase | Build and test platform to VARA specifications. Third-party security audit. Smart contract audit. Penetration testing. Sandbox testing period. |
| 4 | Full License | Final review and approval. VARA issues activity-specific license. Platform authorized for commercial operations. Cooperation agreement with SCA enables UAE-wide operations. |

### 14.1.2 Required VARA Activity Licenses

Based on platform operations, the following VARA activity-specific licenses are required: VA Issuance Services (for TCT and CCT token minting), VA Transfer and Settlement Services (for trade settlement and token transfers), VA Custody Services (for custodial wallet management), and VA Lending and Borrowing Services (for trade financing through liquidity pool). Each activity has specific capital requirements, governance obligations, and ongoing reporting duties defined in VARA's activity-specific rulebooks.

## 14.2 AML/CFT Framework

The platform implements a comprehensive Anti-Money Laundering and Countering the Financing of Terrorism (AML/CFT) framework aligned with UAE Federal AML Law (Federal Decree-Law No. 20 of 2018, as amended) and VARA's specific requirements.

### 14.2.1 Key Controls

**Customer Due Diligence (CDD):** KYC verification for all users before any transaction. Enhanced Due Diligence (EDD) for high-risk users, PEPs, and transactions exceeding $50,000.

**Ongoing Monitoring:** Continuous transaction monitoring against predefined rules (unusual amounts, frequency, geographic patterns). Daily sanctions screening against updated lists (OFAC, UN, EU, UAE local).

**Suspicious Transaction Reporting:** Automated STR generation when monitoring rules trigger. Compliance officer review before GOAML submission. 24-hour reporting deadline for confirmed suspicious activity.

**Record Keeping:** All CDD records retained for minimum 5 years after business relationship ends. Transaction records retained for minimum 5 years after transaction. Records available for regulatory inspection within 48 hours.

**Travel Rule:** Compliance with FATF Travel Rule for virtual asset transfers. Originator and beneficiary information transmitted with all transfers exceeding 3,500 AED (approximately $1,000).

## 14.3 Data Protection

The platform complies with UAE Federal Decree-Law No. 45 of 2021 on Personal Data Protection and DIFC Data Protection Law (where applicable).

**Data Minimization:** Collect only data necessary for platform operations and regulatory compliance. No collection of biometric data beyond what KYC provider requires.

**Consent Management:** Explicit consent obtained for data processing at registration. Granular consent for optional data sharing. Right to withdraw consent (with regulatory exemptions for financial records).

**Cross-Border Transfers:** PII and financial data stored within UAE. Cross-border transfer only with adequate protection or regulatory approval. Cloud infrastructure configured for UAE data residency.

**Breach Notification:** Data breach notification to UAE Data Office within 72 hours. Affected user notification without undue delay. Incident response plan documented and tested annually.

# 15. Security Architecture

Security architecture follows the principle of defense in depth: multiple overlapping security controls ensure no single point of failure can compromise the platform. The guiding philosophy is that complexity is the enemy of security, leading to the use of battle-tested, auditable components over novel but unproven technologies.

## 15.1 Security Layers

| Layer | Controls |
|---|---|
| **Network** | VPC with private subnets for application tier. WAF (AWS WAF) with OWASP Core Rule Set. DDoS protection (AWS Shield Standard). IP whitelisting for admin access and blockchain node communication. mTLS for inter-service communication. |
| **Application** | Input validation on all API endpoints. Output encoding to prevent XSS. CSRF protection via SameSite cookies + anti-CSRF tokens. Rate limiting (100 req/min standard, 10 req/min auth endpoints). Content Security Policy headers. |
| **Data** | AES-256 encryption at rest (RDS TDE, S3 SSE-KMS). TLS 1.3 in transit. Field-level encryption for PII (separate KMS key). Database access via connection pooling with least-privilege roles. Query parameterization to prevent SQL injection. |
| **Blockchain** | Permissioned Avalanche Subnet (no public access). AccessControl.sol enforces role-based function permissions. ReentrancyGuard on all state-changing functions. Pausable contracts for emergency stop. UUPS proxy for secure upgrades with timelock. |
| **Identity** | JWT RS256 with 15-min access tokens. TOTP MFA for admin and high-value operations. Session management with idle timeout (30 min) and absolute timeout (8 hours). Account lockout after 5 failed attempts (30-min cooldown). |
| **Operational** | Principle of least privilege for all service accounts. Infrastructure as Code (Terraform) for auditable deployments. CI/CD pipeline with SAST, DAST, and dependency scanning. Secrets management via AWS Secrets Manager (no hardcoded credentials). |

## 15.2 HSM Key Management

Blockchain private keys are the most critical security asset. All private keys are generated, stored, and used exclusively within AWS CloudHSM (FIPS 140-2 Level 3 certified). Private key material never leaves the HSM boundary.

**Key Generation:** Keys generated inside HSM using FIPS-approved random number generator. BIP-44 hierarchical deterministic derivation for user wallet addresses. Master seed never exported.

**Transaction Signing:** All blockchain transactions signed inside HSM. Application sends unsigned transaction payload to HSM; HSM returns signed transaction. Signing operations logged with transaction hash, timestamp, and requesting service.

**Key Backup:** HSM cluster replication across availability zones. Encrypted backup to separate HSM in DR region. Recovery requires M-of-N quorum of key custodians (3-of-5 for production).

**Key Rotation:** Application API keys: 90-day rotation. HSM operational keys: annual rotation. HSM master key: rotation on security event or personnel change. All rotations logged and auditable.

## 15.3 Smart Contract Security

Smart contracts follow a security-first development methodology using battle-tested OpenZeppelin components.

| Control | Implementation |
|---|---|
| **Access Control** | OpenZeppelin AccessControl with granular roles: MINTER_ROLE, BURNER_ROLE, PAUSER_ROLE, UPGRADER_ROLE, ADMIN_ROLE. Role assignment requires multi-sig approval for production. |
| **Reentrancy Protection** | ReentrancyGuard applied to all functions that modify state and transfer value. Follows checks-effects-interactions pattern throughout. |
| **Emergency Stop** | Pausable contract on all 9 smart contracts. Pause triggered by: admin action, circuit breaker (oracle deviation >5%), or automated anomaly detection. Unpause requires multi-sig. |
| **Upgrade Safety** | UUPS proxy pattern (not transparent proxy) for gas efficiency. Upgrade requires UPGRADER_ROLE + timelock delay (48 hours for production). Storage layout validated by OpenZeppelin upgrade plugin. |
| **Audit** | Pre-mainnet external audit by certified blockchain security firm (CertiK, Trail of Bits, or equivalent). All critical/high findings resolved before deployment. Medium findings documented with mitigation timeline. |

## 15.4 Audit & Logging

Comprehensive audit logging ensures full traceability of all platform operations for regulatory compliance and security investigations.

**What is logged:** Every user action (login, transaction, configuration change), every system event (deployment, scaling, error), every admin operation (user management, KYC review, trade intervention), every blockchain transaction (hash, block, gas, status).

**Log Format:** Structured JSON logs with: timestamp (UTC), actor (user ID or service name), action (verb), target (resource affected), outcome (success/failure), IP address, session ID, correlation ID for distributed tracing.

**Retention:** Application logs: 1 year hot storage (CloudWatch), 7 years cold storage (S3 Glacier) per UAE financial regulation requirements. Blockchain data: permanent (immutable on-chain).

**Tamper Evidence:** Daily log hash anchored to blockchain. Any modification to off-chain logs detectable by hash comparison. Log access restricted to compliance and security roles with audit trail on log access itself.

# 16. Failure Cases & Mitigation Strategies

This section catalogues all identified failure scenarios across the platform, organized by system domain. Each failure includes severity classification, business impact assessment, and detailed mitigation strategy. These scenarios inform the testing strategy (SOW Section 19), risk register (SOW Section 2.4), and default handling framework (SOW Section 9).

## 16.1 Gold Provider Flow — Failure Cases

| ID | Failure Scenario | Severity | Impact | Mitigation Strategy |
|---|---|---|---|---|
| GF-01 | KYC verification fails — Shufti Pro rejects provider identity documents | HIGH | Provider cannot onboard. Potential revenue loss. Provider may abandon platform. | Allow 3 retry attempts with specific rejection reasons. Provide manual KYC escalation path to Compliance Officer. Offer video KYC fallback for document quality issues. |
| GF-02 | Custodian does not respond to vault confirmation email within SLA (5 business days) | CRITICAL | Gold deposit stuck in PENDING_VERIFICATION indefinitely. Provider funds locked without yield. | Auto-escalation after 3 days to Admin. Secondary custodian contact attempt at Day 4. Auto-reject with notification at Day 7. Provider can resubmit with different custodian reference. |

| | | | | |
|---|---|---|---|---|
| **GF-03** | Assay Report shows purity below platform minimum threshold (e.g. <995 fineness) | **MEDIUM** | Gold cannot be tokenized at stated value. Incorrect TCT minting if not caught. | Hard validation rule: reject deposits below 995 fineness. Platform displays minimum purity requirements during deposit form. Admin review catches discrepancies before tokenization. |
| **GF-04** | Gold price oracle returns stale data (LBMA feed delayed >30 minutes) | **CRITICAL** | TCT minting at incorrect valuation. Pool USD-equivalent calculation wrong. Providers over/under credited. | Circuit breaker halts all TCT minting when oracle staleness >30 min. Fallback to last known good price with staleness flag. Admin notification for manual price override. All pending mints queued until oracle recovers. |
| **GF-05** | Insurance certificate expired at time of deposit | **HIGH** | Gold assets not insured against loss/theft. Platform liability exposure. | Automated expiry date validation on upload. Reject deposits with expired insurance. Calendar-based alerts 30 days before existing insurance expiry for active deposits. Require re-upload of valid certificate. |
| **GF-06** | Provider attempts withdrawal but gold is locked as collateral in active trades | **HIGH** | Provider cannot access their assets. Frustration and potential legal dispute. | Real-time display of locked vs available balance. FIFO withdrawal queue with estimated wait time. Partial withdrawal allowed for unlocked portion. Clear messaging showing which trades lock their collateral. |
| **GF-07** | Duplicate gold deposit — same gold registered by two different providers or on two platforms | **CRITICAL** | Double-counting in liquidity pool. Proof of Reserves integrity compromised. | Unique asset fingerprint (custodian + vault + certificate number + weight). Duplicate detection on registration. Cross-reference check during admin verification. Custodian confirmation serves as de-facto uniqueness proof. |
| **GF-08** | Smart contract registerGold() transaction reverts on-chain | **HIGH** | Custody verified off-chain but tokenization fails. Inconsistent state between off-chain and on-chain. | Retry mechanism with exponential backoff (3 attempts). Admin alert on persistent failure. Manual re-trigger capability in Admin Console. All off-chain state held in PENDING until on-chain confirmation received. |

## 16.2 Cash Provider Flow — Failure Cases

| ID | Failure Scenario | Severity | Impact | Mitigation Strategy |
|---|---|---|---|---|
| CF-01 | Wire transfer reference number mismatch — Open Finance API cannot match incoming wire to deposit request | CRITICAL | Provider sent money but system does not recognize it. Manual reconciliation required. Provider funds in limbo. | Fuzzy matching on wire reference (partial match alert). Admin manual reconciliation queue. Provider self-service wire reference correction. Auto-match by amount + sender name + date if reference fails. 48-hour SLA for manual resolution. |
| CF-02 | AML screening flags provider — GOAML check returns suspicious activity indicator | CRITICAL | Deposit frozen. Regulatory reporting obligation triggered. Potential legal exposure. | Immediate deposit freeze (no CCT minting). Auto-generate SAR (Suspicious Activity Report) draft for Compliance. Escalation to Compliance Officer within 4 hours. Separate from provider communication (tipping-off prevention). Wire refund only after compliance clearance. |
| CF-03 | Provider sends wrong amount — wire amount does not match deposit request | MEDIUM | Amount mismatch blocks automatic processing. Over-payment or under-payment creates reconciliation issue. | Tolerance threshold: accept within +/- 1% and adjust CCT minting accordingly. Outside threshold: hold for admin review. Provider notified of mismatch with correction options (top-up wire or accept adjusted amount). Refund process for over-payment. |
| CF-04 | Open Finance Banking API downtime — cannot confirm wire receipt | HIGH | All cash deposits blocked. Providers waiting indefinitely for confirmation. | Health check monitoring with 5-min intervals. Admin manual confirmation capability using bank statement upload. Queue deposits during outage, auto-process when API recovers. Provider notification of delay with estimated resolution. |
| CF-05 | Withdrawal requested but insufficient pool liquidity (all funds locked in escrow) | HIGH | Provider cannot access their money. Trust erosion. Potential run on the pool if communicated poorly. | FIFO withdrawal queue with transparent position tracking. Estimated processing date based on trade repayment schedule. Partial withdrawal up to available amount. Dashboard showing pool utilization and locked amounts. 15% reserve ratio maintained to minimize this scenario. |
| CF-06 | CCT mint transaction fails on-chain after wire confirmed | HIGH | Wire received and confirmed but tokens not issued. Inconsistent state. | Same pattern as GF-08: retry with backoff, admin alert, manual re-trigger. Off-chain status stays WIRE_RECEIVED until on-chain mint confirmed. Provider sees accurate status throughout. |

## 16.3 Trade Financing Flow — Failure Cases

| ID | Failure Scenario | Severity | Impact | Mitigation Strategy |
|---|---|---|---|---|
| TF-01 | AI document verification flags trade documents as potentially fraudulent or inconsistent | CRITICAL | Trade application blocked. Buyer may be submitting forged documents. Financial loss if undetected. | Auto-reject with detailed discrepancy report to Admin. Buyer receives generic rejection (no fraud detection details disclosed). Compliance officer deep review with original document request. Repeat offenders flagged in system. Escalation to legal if pattern detected. |
| TF-02 | Credit scoring returns zero — new buyer with no trade history (cold-start problem) | MEDIUM | System cannot assess risk. Auto-rejection would block all initial pilot trades. | MVP cold-start strategy: Admin manual override with documented rationale. Reduced financing limit (50% of requested). Higher collateral requirement. Mandatory senior admin approval for first 3 trades per new buyer. Build history for rules-based scoring. |
| TF-03 | Trade financing amount exceeds 25% pool concentration limit | HIGH | Deal cannot be approved despite creditworthy buyer. Large deals blocked. | Pre-check during application showing max available financing. Admin option to split deal into tranches. Queue mechanism for upcoming pool capacity. Alternative: partial financing with buyer covering remainder. Dashboard showing real-time pool concentration. |
| TF-04 | Buyer defaults on repayment — misses payment deadline | CRITICAL | Provider funds at risk. Platform revenue impacted. Pool liquidity reduced. | 4-phase Default Handling Framework (SOW Section 9): Grace Period (7 days, late fee accrues) > Late Payment (Day 8-30, penalty escalation) > Pre-Default (Day 31-60, formal notice) > Default Declaration (Day 61+, recovery waterfall). 5-priority recovery: penalties > reserve fund > insurance > pro-rata provider > legal action. |
| TF-05 | Supplier fails to deliver goods after milestone 1 payment released (30%) | CRITICAL | Buyer has no goods. 30% of financing already released to supplier. Escrow still holds remaining 70%. | Remaining milestones frozen immediately. Dispute Resolution Contract initiated (SOW Section 3, Journey 4). Escrow vault holds remaining 70% until resolution. Evidence submission window for both parties. Admin-led arbitration with 4 possible outcomes: full refund, partial settlement, milestone restructure, or trade cancellation with penalty. |
| TF-06 | Milestone documentation incomplete — supplier uploads partial shipping docs | MEDIUM | Milestone approval delayed. Supplier payment delayed. Buyer cannot confirm receipt. | Document completeness checklist enforced at upload (mandatory fields). Clear rejection with specific missing documents listed. Supplier gets 2 retry attempts before admin escalation. Buyer not asked to confirm until docs are complete. |
| TF-07 | Escrow lockFunds() fails — insufficient pool balance at execution time (race condition) | HIGH | Trade approved but cannot be funded. Inconsistent state between approval and escrow. | Atomic check-and-lock pattern in smart contract. Pre-reservation of funds during approval (soft lock). If reservation expires (24h), deal returns to approval queue. Real-time pool balance check at both approval and escrow stages. |
| TF-08 | Buyer disputes quality of goods after confirming milestone receipt | MEDIUM | Buyer wants reversal after on-chain release. Funds already with supplier. | Confirmation is binding — clearly stated in platform T&Cs. Post-confirmation disputes handled through Dispute Resolution (off-chain arbitration for released funds). Insurance mechanism for quality disputes. Time-limited confirmation window (48h to review before auto-confirm). |

| | | | | |
|---|---|---|---|---|
| **TF-09** | Chill Pay off-ramp fails — supplier cannot convert tokens to fiat | **HIGH** | Supplier has tokens but cannot use them. Defeats purpose of platform. | Retry mechanism with Chill Pay (3 attempts). Admin fallback: manual bank wire from platform operational account. Alternative off-ramp path (direct bank transfer using Open Finance API). Supplier notified with estimated resolution time. Tokens remain in wallet until successful off-ramp. |
| **TF-10** | Trade NFT minting fails after admin approval | **HIGH** | Deal approved off-chain but not recorded on-chain. No escrow can be created. | Retry with exponential backoff. Admin re-trigger capability. All downstream steps blocked until Trade NFT confirmed. Off-chain status clearly shows PENDING_CHAIN_CONFIRMATION . Gas estimation pre-check before mint attempt. |

## 16.4 Blockchain & Smart Contract Failures

| ID | Failure Scenario | Severity | Impact | Mitigation Strategy |
|---|---|---|---|---|
| **BF-01** | Avalanche Subnet validator node goes offline (1 of 4-5 validators) | **HIGH** | Reduced consensus capacity. If >1/3 validators down, chain halts. No transactions processed. | Minimum 4 validators across different availability zones. Health monitoring with 1-min alerting. Auto-restart scripts on validator nodes. DevOps runbook for manual failover. 15-minute RTO target for single node recovery. |
| **BF-02** | Smart contract bug discovered after deployment (e.g. calculation error in Yield Engine) | **CRITICAL** | Incorrect yield distribution. Potential financial loss for providers. Trust breach. | UUPS upgrade proxy pattern allows contract upgrade without redeployment. Emergency Pausable modifier halts affected contract immediately. Admin triggers pause via Admin Console. Fix deployed through proxy upgrade after audit. Post-incident: recalculate and compensate affected providers. |
| **BF-03** | Reentrancy attack attempt on Escrow Vault | **CRITICAL** | Funds drained from escrow through malicious contract interaction. | ReentrancyGuard (OpenZeppelin) on all external calls. Checks-Effects-Interactions pattern enforced. Private permissioned subnet limits attack surface (only whitelisted addresses). Regular Slither/Mythril scans. Pre-deployment audit covers reentrancy vectors. |
| **BF-04** | Gas estimation failure — complex transaction exceeds block gas limit | **MEDIUM** | Transaction reverts. Operation (mint, transfer, milestone release) fails. | Gas estimation pre-check before submission. Batch operations split into smaller transactions. Subnet gas limit configured for expected transaction complexity. Admin monitoring of gas usage patterns. |
| **BF-05** | Private key compromise — deployer or admin key leaked | **CRITICAL** | Attacker can upgrade contracts, pause system, drain funds, or manipulate access control. | AWS KMS key management (no keys on disk). Multi-sig requirement for critical operations (upgrade, pause, role grant). Key rotation every 90 days. Immediate revocation capability. AccessControl role separation (deployer vs operator vs admin). Incident response plan with <15 min key revocation SLA. |
| **BF-06** | Chain reorganization or fork in subnet | **LOW** | Transaction history inconsistency. Potential double-spend. | Private permissioned subnet with controlled validators makes reorg extremely unlikely. Avalanche consensus provides sub-2s finality. Application layer waits for finality before confirming operations. Monitoring for unusual finality delays. |

## 16.5 External Integration Failures

| ID | Failure Scenario | Severity | Impact | Mitigation Strategy |
|---|---|---|---|---|
| EF-01 | Shufti Pro API downtime — KYC service unavailable | HIGH | All new user registrations blocked. Onboarding pipeline halted. | API health monitoring with 5-min checks. Queue registration requests during downtime. Admin manual KYC approval as fallback (with enhanced documentation). SLA requirement from Shufti Pro in contract (99.5% uptime). Process queued requests automatically on recovery. |
| EF-02 | Chill Pay integration failure — payment gateway returns errors | HIGH | Suppliers cannot off-ramp tokens to fiat. Blocked revenue realization. | Retry mechanism (3 attempts with 5-min intervals). Admin manual wire transfer as fallback. Token balance preserved in supplier wallet until successful off-ramp. Real-time status tracking with estimated resolution. Chill Pay SLA escalation process. |
| EF-03 | LBMA gold price feed complete outage (>1 hour) | CRITICAL | No gold valuation available. All gold-related operations halted (deposits, TCT minting, pool valuation). | Circuit breaker triggers at 30-min staleness. Fallback to secondary feed (if available in future phases). Admin manual price entry with documented source. All gold operations queued (not rejected). Pool valuation frozen at last known price until recovery. Provider dashboard shows staleness warning. |
| EF-04 | Open Finance Banking API returns incorrect wire match (false positive) | CRITICAL | Wrong wire matched to deposit. Provider credited for someone else's wire. Financial loss. | Multi-field matching required (reference + amount + sender). Admin review mandatory before CCT minting. Daily reconciliation report comparing API matches vs bank statements. Automatic mismatch detection alerts. 4-hour reversal window before minting. |
| EF-05 | GOAML reporting system unavailable during mandatory filing deadline | HIGH | Regulatory non-compliance. Potential fines from UAE FIU. | Local queue of pending reports with auto-retry. Compliance officer manual filing via GOAML web portal. Audit log preserves all report data regardless of submission status. Regulatory notification of technical difficulty if deadline missed. |
| EF-06 | Custodian changes vault location without notifying platform | MEDIUM | On-chain metadata does not match actual gold location. Proof of Reserves inaccurate. | Quarterly custodian confirmation requirement. Automated reminder emails to custodians. Annual audit includes physical vault verification. Any metadata change triggers re-verification workflow. |

## 16.6 Security & Compliance Failures

| ID | Failure Scenario | Severity | Impact | Mitigation Strategy |
|---|---|---|---|---|
| **SF-01** | Unauthorized access — user gains admin privileges through session hijack or privilege escalation | **CRITICAL** | Attacker can approve deals, release escrow funds, modify user roles. Complete system compromise. | MFA enforcement on all admin accounts. JWT token with short expiry (15 min) + refresh token rotation. IP whitelisting for Admin Console. Session anomaly detection (geo, device). AccessControl on-chain prevents unauthorized smart contract calls regardless of off-chain compromise. |
| **SF-02** | Data breach — PII (KYC documents, trade data) exposed | **CRITICAL** | PDPL (UAE data protection) violation. Fines up to AED 10M. Reputation destruction. | PII never stored on-chain (only metadata hashes). AES-256 encryption at rest for all off-chain PII. TLS 1.3 in transit. UAE data residency (AWS Middle East region). Database access logging and anomaly detection. Incident response plan with 72-hour notification to PDPL authority. |
| **SF-03** | VARA regulatory audit finds non-compliance in token operations | **HIGH** | Operating license risk. Fines up to AED 10M for marketing violations. Potential platform shutdown order. | Quarterly internal compliance audit. All token operations logged in immutable audit trail. Complete KYC/AML documentation for every participant. Legal counsel review before any public marketing. VARA Rulebook compliance checklist maintained by Compliance Officer. |
| **SF-04** | Denial of Service attack on platform API endpoints | **MEDIUM** | Platform unavailable. Users cannot access portals. Trades stalled. | Rate limiting on all API endpoints. CDN with DDoS protection (CloudFlare/AWS Shield). Auto-scaling backend infrastructure. On-chain operations unaffected by off-chain DDoS. Emergency maintenance page with status updates. |
| **SF-05** | Insider threat — platform employee manipulates trade approvals or yield calculations | **CRITICAL** | Financial fraud. Provider losses. Legal liability. Trust destruction. | Separation of duties (no single person can approve + fund). All admin actions logged in immutable audit trail. Multi-approval for high-value operations (>$100K). Smart contract enforces rules regardless of admin intent. Quarterly audit of admin action logs. Whistleblower policy. |

# 17 Edge Cases — High Level

Edge cases represent unusual but plausible scenarios that fall outside the normal happy-path flow. These must be considered during design and development to prevent unexpected system behavior, financial discrepancies, or user confusion.

## 17.1 Liquidity Pool Edge Cases

| ID | Edge Case | Why It Matters | Handling Approach |
|---|---|---|---|
| EC-01 | Gold price drops 20%+ in a single day (flash crash) | TCT valuation drops suddenly. Pool USD-equivalent shrinks. Active trade financing may become under-collateralized. Provider LP share value drops. | Circuit breaker pauses new TCT minting during >10% daily swing. Active trades unaffected (locked at deal-time valuation). Providers notified of valuation change. Admin review of pool health metrics triggered. |
| EC-02 | All providers request simultaneous withdrawal (bank run scenario) | Pool drained completely. Active trade escrows at risk. Platform cannot function. | 15% reserve ratio provides buffer. FIFO queue ensures fair ordering. Active escrow funds are contract-locked and cannot be withdrawn. Platform communicates transparent liquidity status. Emergency: temporary withdrawal cooldown (72h) with admin approval. |
| EC-03 | Single provider owns >80% of the pool | Concentration risk. Single withdrawal destabilizes entire pool. Disproportionate yield distribution. | Dashboard shows provider concentration metrics. Alert when single provider >50% of pool. Consider maximum single-provider cap (e.g., 60%). Encourage diversification in provider onboarding. |
| EC-04 | Pool utilization reaches 100% (all funds locked in escrow) | No new trades can be funded. No withdrawals possible. Platform appears non-functional. | Dashboard shows real-time utilization with traffic-light indicator. Auto-pause new trade approvals at 85% utilization. 15% reserve ratio prevents true 100%. Admin notification at 75%, 85%, 95% thresholds. |
| EC-05 | Zero trades for extended period (no yield generation) | Providers earn zero yield. Platform earns zero revenue. Providers may withdraw, creating doom loop. | Minimum guaranteed yield consideration for Phase 2. Transparent communication that yield depends on trade volume. Provider dashboard shows projected vs actual APY. Active trade pipeline visibility for providers. |

## 17.2 Trade Lifecycle Edge Cases

| ID | Edge Case | Why It Matters | Handling Approach |
|---|---|---|---|
| EC-06 | Buyer and supplier are related parties (same beneficial owner) | Potential collusion for fraudulent trade financing. Self-dealing risk. | KYC captures UBO (Ultimate Beneficial Owner) for all parties. Cross-reference UBO database during trade application. Admin review flag when UBO overlap detected. Mandatory enhanced due diligence for related-party trades. |
| EC-07 | Trade commodity is perishable and expires during milestone delays | Goods worthless by delivery. Buyer refuses acceptance. Supplier payment disputed. | Commodity type classification during application. Perishable goods flagged for expedited milestone processing. Shorter milestone SLAs for perishable trades. Insurance requirement for perishable commodities. Force-close mechanism if milestone delayed beyond commodity shelf life. |
| EC-08 | Milestone 1 approved but Milestone 2 rejected, then Milestone 3 submitted | Non-sequential milestone completion. Partial delivery scenario. | Milestones must be sequential (M2 blocked until M1 released). Rejected milestones require resubmission before next milestone. Clear UI showing milestone dependency chain. Admin override for exceptional circumstances with documented rationale. |
| EC-09 | Buyer pays back early (before scheduled repayment date) | System must handle early repayment correctly. Yield calculation adjustment needed. | Early repayment allowed and encouraged. Yield calculated pro-rata up to actual repayment date. No early repayment penalty in MVP. Pool replenished faster, benefiting all providers. Trade NFT updated to COMPLETED early. |

| | | | |
|---|---|---|---|
| **EC-10** | Trade currency fluctuation — deal structured in USD but buyer repays in AED | Exchange rate difference between deal creation and repayment. Who absorbs FX risk? | Deal denomination fixed at creation (USD or AED, not mixed). Chill Pay handles conversion at prevailing rate. FX risk disclosure in trade terms. Platform does not absorb FX risk — clearly stated in T&Cs. Future: FX hedging option in Phase 2. |
| **EC-11** | Same supplier appears in multiple active trades exceeding platform exposure limits | Supplier default impacts multiple trades simultaneously. Correlated risk. | Supplier exposure tracking across all active trades. Aggregate supplier limit (e.g., max 40% of pool exposure per supplier). Admin alert when supplier approaches limit. New trade applications blocked when limit reached for that supplier. |

## 17.3 System & Operational Edge Cases

| ID | Edge Case | Why It Matters | Handling Approach |
|---|---|---|---|
| **EC-12** | Yield Engine daily run fails mid-calculation (partial distribution) | Some providers receive yield, others do not. Inconsistent state. | Atomic yield distribution — all-or-nothing per daily run. Failed runs retry automatically (3 attempts). Admin notification on failure. Manual trigger capability. Rollback mechanism for partial distributions. Audit log captures every calculation step. |
| **EC-13** | Multiple admins approve the same trade simultaneously (race condition) | Duplicate approvals. Double escrow funding. Financial discrepancy. | Optimistic locking on trade approval (first approval wins). Database-level unique constraint on trade status transition. UI real-time status refresh. Second approver sees already-approved status. Smart contract rejects duplicate escrow lock attempts. |
| **EC-14** | User has active trades and active deposits when account is suspended (compliance issue) | Cannot simply freeze account — active trades and provider deposits have counterparties. | Account suspension freezes new operations only. Active trades continue to completion (counterparty protection). Active deposits remain in pool but new deposits blocked. Withdrawal request queued for compliance review. Clear separation between account freeze and asset freeze. |
| **EC-15** | Platform upgrade or maintenance during active trade milestone window | Trades have time-sensitive milestones. Downtime could cause milestone SLA breaches. | Planned maintenance windows communicated 72h in advance. Milestone SLA timers paused during scheduled maintenance. On-chain operations unaffected by off-chain maintenance. Emergency maintenance: admin extends milestone deadlines. Rolling deployment strategy to minimize downtime. |
| **EC-16** | Leap second, timezone change, or daylight saving impacts daily yield calculation | Yield Engine timing discrepancy. Double-run or missed run. | All timestamps in UTC (no timezone dependencies). Yield Engine uses block timestamps (on-chain) not server time. Daily run triggered by scheduler with idempotency check. Duplicate run detection and prevention. Audit trail includes exact calculation timestamp. |
| **EC-17** | Provider completes KYC but Shufti Pro later discovers fraudulent documents (post-verification fraud) | Verified provider operating on platform with fraudulent identity. Regulatory breach. | Shufti Pro ongoing monitoring alerts. Immediate account suspension on post-KYC fraud alert. Active deposits frozen pending investigation. GOAML suspicious activity report filed. All trades involving provider flagged for admin review. DIDVerifier.revokeAccess() called on-chain. |

# 18. Module Breakdown & Dependencies

The platform is decomposed into 15 modules (originally 13 in v3.0, with M5 Backend split into 4 sub-modules for delegation clarity). Each module is scoped for assignment to an individual engineer or pair. Module dependencies define the critical path and dictate build sequence.

## 18.1 Module Registry

| ID | Module | Scope | Depends On |
|---|---|---|---|
| **M1** | **Blockchain Infrastructure** | Avalanche Subnet deployment, validator setup, genesis configuration, RPC endpoints, block explorer | None - foundational. Must be complete before any on-chain module. |
| **M2** | **Smart Contracts** | All 9 Solidity contracts: AssetTokenFactory, LiquidityPool, EscrowVault, MilestonePayment, TradeNFT, InvoiceNFT, DIDVerifier, CreditScore, YieldDistributor | M1 (deployed subnet for testing) |
| **M3** | **HSM & Key Management** | AWS CloudHSM setup, key derivation (BIP-44), signing service, key rotation, backup/recovery | M1 (blockchain addresses needed) |
| **M4** | **Oracle Service** | LBMA price feed integration, on-chain price updates, circuit breaker, staleness detection, fallback sources | M1, M2 (PriceOracle contract) |
| **M5a** | **Backend: IAM & Auth** | User registration, authentication (JWT + MFA), role management, Shufti Pro KYC integration, wallet provisioning trigger, session management | M3 (wallet provisioning needs HSM) |
| **M5b** | **Backend: Provider Service** | Gold deposit workflow, cash deposit workflow, withdrawal processing, FIFO queue, yield tracking, provider portfolio calculations | M2 (token contracts), M5a (auth), M4 (gold pricing) |
| **M5c** | **Backend: Trade & Settlement** | Trade application processing, credit scoring engine, escrow management, milestone workflows, repayment tracking, dispute workflows, Chill Pay integration | M2 (trade contracts), M5a (auth), M5b (pool data) |
| **M5d** | **Backend: Shared Services** | Notification service (email/push), file upload/storage (S3), audit logging, GOAML reporting, scheduled jobs (yield calc, reminders), admin utilities | M5a (auth context) |
| **M6** | **Provider Portal (Frontend)** | Provider registration, deposit/withdrawal flows, portfolio dashboard, yield tracking, withdrawal queue status | M5a, M5b (APIs) |
| **M7** | **Trader Portal (Frontend)** | Trader registration, trade application form, milestone tracking, repayment dashboard, dispute filing UI | M5a, M5c (APIs) |

| | | | |
|---|---|---|---|
| **M8** | **Admin Console (Frontend)** | KYC review, deposit verification, trade approval, dispute management, compliance dashboard, system settings, user management | M5a, M5b, M5c, M5d (all APIs) |
| **M9** | **Compliance Dashboard** | Proof of Reserves display, risk metrics, transaction monitoring, AML alerts, GOAML exports, regulatory observer view | M5d (audit/compliance APIs), M2 (on-chain data) |
| **M10** | **AI/ML Service** | Document verification (invoice authenticity, consistency checks), OCR processing. MVP: Rule-based with confidence scoring. Phase 1: ML model training from pilot data. | M5c (feeds documents for verification) |
| **M11** | **Yield Engine** | Daily yield calculation, monthly distribution trigger, LP value computation, distribution history, provider APY projections | M2 (YieldDistributor contract), M5b (pool data) |
| **M12** | **DevOps & Infrastructure** | AWS me-south-1 setup, CI/CD pipelines, monitoring (Grafana/Prometheus), log aggregation, backup strategy, disaster recovery | Parallel with all modules from Phase 1 |
| **M13** | **Testing & Audit** | Smart contract test suite (>90% coverage), integration tests, load testing (k6), security testing, external audit coordination, UAT support | All modules (testing is continuous from Phase 2) |

## 18.2 Module Dependency Chain

PHASE 1 (Foundation): M1 > M12 (parallel). PHASE 2 (Core): M2 + M3 (parallel, both need M1). PHASE 3 (Services): M4 + M5a (parallel) > M5b + M5c + M5d + M10 + M11 (parallel, all need M5a). PHASE 4 (UI + Integration): M6 + M7 + M8 + M9 (parallel frontend builds). PHASE 5 (Quality): M13 (testing across all modules). The critical path runs: M1 > M2 > M5a > M5c > M7/M8 > M13.

# 19. Delivery Roadmap

The project is delivered across 5 phases with clear milestones and go/no-go gates between each phase. Each phase produces demonstrable, testable deliverables. Total estimated timeline: 12-15 months depending on team size and external dependency resolution.

## 19.1 Phase Overview

| Phase | Duration | Deliverables | Go/No-Go Criteria |
|---|---|---|---|
| **Phase 1: Foundation** | Months 1-2 | Avalanche Subnet live with 4 validators. AWS infrastructure provisioned (me-south-1). CI/CD pipelines operational. HSM configured. Dev/staging/prod environments ready. | Subnet produces blocks. Validator nodes sync. HSM generates keys. All environments accessible. Monitoring dashboards live. |
| **Phase 2: Smart Contracts** | Months 2-4 | All 9 smart contracts deployed to testnet. >90% test coverage. Oracle service feeding gold prices. Basic token minting/burning verified. Escrow lock/release tested. | All contracts pass test suite. No critical/high vulnerabilities in internal review. Oracle price updates working. Gas optimization complete. |
| **Phase 3: Backend + API** | Months 3-7 | All backend services operational (M5a-d). KYC integration working. Banking API connected. AI document verification running. Credit scoring engine operational. Yield calculation verified. | End-to-end deposit flow works (gold + cash). Trade application processes through all steps. Milestone payment triggers work. API documentation complete. |
| **Phase 4: Frontend + Integration** | Months 6-10 | All 4 portals functional (Provider, Trader, Admin, Compliance). Chill Pay integration tested. Full end-to-end flows working. UI/UX polished. Responsive design verified. | Complete user journey works: Provider deposits > Trader applies > Admin approves > Milestones release > Settlement complete. All portals pass UAT. |
| **Phase 5: Testing + Launch** | Months 9-12+ | External smart contract audit. Load testing (100 concurrent users, 50 simultaneous trades). Security penetration testing. Bug fixes from audit. UAT with pilot counterparties. Production deployment. | Audit report: 0 Critical, 0 High. Load test passes all NFR thresholds. Security test clear. UAT signed off by Gary. VARA compliance verified. Pilot Deal 1 executed. |

## 19.2 Key Milestones

| ID | Milestone | Description | Target |
|---|---|---|---|
| **MS-1** | Subnet Live | Avalanche Evergreen Subnet producing blocks with 4 validators | End of Month 2 |
| **MS-2** | Smart Contracts Deployed | All 9 contracts on testnet with >90% test coverage | End of Month 4 |
| **MS-3** | First Token Minted | TCT and CCT tokens minted end-to-end on testnet (deposit to token) | Month 5 |
| **MS-4** | First Trade Processed | Full trade lifecycle on testnet: application > approval > escrow > milestones > settlement | Month 7 |

| | | | |
|---|---|---|---|
| **MS-5** | Portal Demo | All 4 portals functional. Demo to Gary with simulated pilot deal | Month 8 |
| **MS-6** | External Audit Complete | Third-party smart contract audit complete with 0 Critical/High findings | Month 10 |
| **MS-7** | UAT Sign-Off | User Acceptance Testing complete with pilot counterparties. Gary signs off. | Month 11 |
| **MS-8** | Pilot Deal 1 Live | First real pilot trade ($5M Food deal) executed on production platform | Month 12-13 |
| **MS-9** | All Pilots Complete | All 3 pilot deals ($16M total) processed. Platform model validated. | Month 14-15 |

# 20. Quality Assurance Scope

Quality assurance is embedded throughout the development lifecycle, not treated as a final phase. Given the financial nature of the platform ($16M pilot, real money), the QA strategy emphasizes automated testing, smart contract security, and rigorous integration testing.

## 20.1 Testing Strategy by Layer

| Layer | Approach | Details |
|---|---|---|
| **Smart Contracts** | Developer-owned + audit | Hardhat test suite with >90% line coverage. Fuzzing with Foundry for edge cases. Formal verification for EscrowVault and LiquidityPool (highest-value contracts). External audit by CertiK or Trail of Bits before mainnet deployment. All tests run in CI on every PR. |
| **Backend APIs** | Developer-owned + QA validation | Unit tests (Jest/Mocha) for business logic. Integration tests for API endpoints. Contract tests for blockchain RPC calls. Database migration tests. QA writes additional scenario-based API tests using Postman/Newman. |
| **Frontend** | Developer unit + QA E2E | Component-level unit tests (React Testing Library). QA owns end-to-end tests using Playwright or Cypress for critical user journeys (deposit flow, trade application, milestone release). Cross-browser testing: Chrome, Firefox, Safari. |
| **Integration** | QA-owned | End-to-end workflow tests covering complete user journeys across frontend > backend > blockchain > external services. Includes: KYC flow, deposit-to-token flow, complete trade lifecycle, yield distribution cycle. Run nightly on staging environment. |
| **Performance** | QA + DevOps | Load testing with k6: target 100 concurrent users, 50 simultaneous active trades. API response time < 200ms (P95). Blockchain transaction throughput >100 TPS. Stress testing at 3x expected load to find breaking points. |
| **Security** | External + internal | OWASP Top 10 verification. Dependency vulnerability scanning (Snyk). Smart contract-specific: reentrancy, integer overflow, access control bypass. External penetration test in Phase 5. Bug bounty program post-launch. |

## 20.2 UAT (User Acceptance Testing)

UAT is conducted in Phase 5 with real pilot counterparties. Gary coordinates participant involvement. UAT scripts are prepared by QA based on the user journeys defined in Section 3. Key UAT scenarios: Provider deposits gold and sees TCT + LP tokens in portal. Provider deposits cash via wire and sees CCT tokens.

Trader applies for financing with real trade documents. Admin approves deal and milestones are created. Supplier receives milestone payment and uses Chill Pay off-ramp. Complete yield cycle runs and providers see returns.

## 20.3 Smart Contract Audit Timeline

| Activity | Timeline & Details |
|---|---|
| **Audit RFP** | Month 7: Send RFP to 3 audit firms (CertiK, Trail of Bits, OpenZeppelin). Include scope: 9 contracts, ~3000 lines Solidity. Select firm by Month 8. |
| **Code Freeze** | Month 8: Smart contract code frozen for audit. No new features, only critical bug fixes allowed during audit period. |
| **Audit Execution** | Months 9-10: 4-6 week audit period. Weekly sync calls with auditors. Immediate triage of critical findings. |
| **Remediation** | Month 10-11: Fix all Critical and High findings. Document Medium findings with mitigation timeline. Re-audit of fixes (typically 1-2 weeks). |
| **Final Report** | Month 11: Published audit report. Zero Critical/High findings = proceed to production. Any remaining = delay until resolved. |

# 21. Change Control Process

SOW v4.0 becomes the frozen baseline upon sign-off. Any changes to scope, features, architecture, or timeline must follow this formal change control process. This protects against scope creep (Risk R-7) on a $2.4M project.

| Step | Details |
|---|---|
| **1. Change Request** | Any stakeholder submits a Change Request Form (CRF) describing: what is changing, why it is needed, affected modules, estimated effort impact, timeline impact, and risk assessment. |
| **2. Impact Assessment** | Technical Lead and Engineering Manager assess: development effort (person-days), module dependencies affected, regression risk, timeline impact. Assessment completed within 3 business days. |
| **3. Approval** | Changes categorized: MINOR (< 2 person-days, no new dependencies) approved by EM. MAJOR (> 2 person-days or new dependencies) requires Gary + Dinesh approval. CRITICAL (architecture/scope change) requires full stakeholder sign-off. |
| **4. Implementation** | Approved changes added to backlog with CRF reference. Sprint scope adjusted. SOW version incremented (v4.1, v4.2 etc.). Change log maintained in Appendix. |
| **5. Verification** | Changed features re-tested against updated acceptance criteria. Regression tests run on affected modules. Stakeholder review of implemented change. |

# 22. Feature-to-Module Traceability Matrix

Every feature from Section 4 maps to exactly one module owner. This matrix ensures zero features fall through cracks during sprint planning and provides clear accountability. If a feature ID does not appear in this matrix, it is unassigned and represents a planning gap.

| Feature | Name | Priority | Module | Notes |
|---|---|---|---|---|
| **IAM-01** | User Registration | P0 | M5a | Backend auth service + Provider/Trader portal UI |
| **IAM-02** | KYC Integration | P0 | M5a | Shufti Pro API integration in backend. UI in respective portals. |
| **IAM-03** | Role-Based Access | P0 | M5a | RBAC middleware. Permission matrix enforced at API gateway. |
| **IAM-04** | MFA Authentication | P0 | M5a | TOTP implementation. Admin MFA mandatory. |
| **IAM-05** | On-Chain DID | P0 | M2 + M5a | Contract in M2. Backend integration in M5a. |
| **IAM-06** | Wallet Provisioning | P0 | M3 + M5a | HSM key gen in M3. Triggered by M5a on registration. |
| **AP-01** | Gold Deposit | P0 | M5b + M6 | Backend flow in M5b. Provider Portal UI in M6. |
| **AP-02** | Cash Deposit | P0 | M5b + M6 | Banking API integration in M5b. UI in M6. |
| **AP-03** | Gold Withdrawal | P0 | M5b + M6 | FIFO queue logic in M5b. UI in M6. |
| **AP-04** | Cash Withdrawal | P0 | M5b + M6 | Same as AP-03 for cash. T+1 wire in M5b. |
| **AP-05** | Provider Dashboard | P1 | M6 | Frontend dashboard consuming M5b APIs. |
| **TM-01** | Trade Application | P0 | M5c + M7 | Backend processing in M5c. Trader Portal UI in M7. |
| **TM-02** | Credit Scoring | P0 | M5c + M10 | Rules engine in M5c. AI scoring in M10. |
| **TM-03** | Milestone Payments | P0 | M2 + M5c | Contract in M2. Backend orchestration in M5c. |
| **TM-04** | Trade NFT | P0 | M2 | Smart contract only. Backend calls via M5c. |
| **TM-05** | Escrow Management | P0 | M2 + M5c | EscrowVault contract in M2. Backend in M5c. |
| **TM-06** | Repayment Tracking | P1 | M5c + M7 | Backend tracking in M5c. Trader Portal UI in M7. |
| **TM-07** | Chill Pay Off-Ramp | P1 | M5c + M7 | Redirect integration in M5c. UI flow in M7. |
| **CR-01** | AML Screening | P0 | M5d + M9 | Screening service in M5d. Dashboard in M9. |
| **CR-02** | Transaction Monitoring | P0 | M5d + M9 | Rule engine in M5d. Alert queue in M9. |
| **CR-03** | Proof of Reserves | P0 | M9 | Dashboard querying on-chain + off-chain data. |
| **CR-04** | Risk Dashboard | P1 | M9 | Compliance dashboard module. Previously unassigned (F-06 fix). |
| **CR-05** | Regulator Observer | P1 | M9 | Read-only portal view within compliance dashboard. |

| | | | | |
|---|---|---|---|---|
| **CR-06** | Audit Trail | P0 | M5d | Logging service in M5d. Previously unassigned (F-06 fix). |
| **YE-01** | Yield Calculation | P0 | M11 | Yield Engine. Feature ID created per F-06. |
| **YE-02** | Distribution Trigger | P0 | M11 + M2 | Backend trigger in M11. On-chain in M2. Feature ID created per F-06. |
| **YE-03** | LP Value Update | P0 | M11 | Computation engine in M11. Feature ID created per F-06. |

# Appendix A: Trade Finance & Platform Glossary

This glossary defines domain-specific terms used throughout this document. Engineers, QA, and stakeholders unfamiliar with trade finance should reference this section when encountering unknown terminology.

| Term | Definition |
|---|---|
| **AML/CFT** | Anti-Money Laundering / Combating the Financing of Terrorism. Regulatory framework requiring financial platforms to detect and prevent illicit financial activity. |
| **ARVA** | Asset-Referenced Virtual Asset. VARA classification for tokens backed by real-world assets (gold, cash). Requires 100% backing ratio at all times. |
| **Assay Report** | Laboratory analysis certificate confirming gold purity (fineness). Must be issued by LBMA-certified assayer. Required for gold deposit verification. |
| **BIP-44** | Bitcoin Improvement Proposal 44. Hierarchical deterministic wallet standard for deriving multiple addresses from a single master key. Used for user wallet generation. |
| **CCT** | Cash Collateral Token. ERC-20 token representing verified cash deposits. 1 CCT = 1 USD. Minted when banking API confirms wire receipt. |
| **DIFC** | Dubai International Financial Centre. Free zone with independent regulatory framework. DIFC Innovation Testing License serves as VARA sandbox alternative. |
| **ERC-20/721** | Ethereum Request for Comment standards. ERC-20: fungible token standard (TCT, CCT, LP). ERC-721: non-fungible token standard (TradeNFT, InvoiceNFT). |
| **Ex Works** | Incoterm where seller makes goods available at their premises. Buyer assumes all transport costs and risks from that point. Common in commodity trades. |
| **FSSAI** | Food Safety and Standards Authority of India. Regulatory body whose certification may be required for food commodity imports from India. Relevant to Food vertical pilot. |
| **GOAML** | GoAML (Anti-Money Laundering system). UAE Financial Intelligence Unit's platform for receiving Suspicious Transaction Reports (STRs) from financial institutions. Integration required for VARA compliance. |
| **HSM** | Hardware Security Module. Tamper-resistant hardware for cryptographic key management. AWS CloudHSM (FIPS 140-2 Level 3) used for all platform wallet keys. |
| **ICPO** | Irrevocable Corporate Purchase Order. Formal commitment from buyer to purchase specified goods. One of the trade documents uploaded during financing application. |
| **LBMA** | London Bullion Market Association. Sets the global benchmark gold price (AM and PM fix) used by central banks and institutions. Primary oracle source for TCT valuation. |
| **LOI** | Letter of Intent. Non-binding document expressing buyer's intention to purchase. Early-stage trade document in the deal formation process. |
| **LP Token** | Liquidity Provider Token. ERC-20 token representing proportional share of the unified liquidity pool. Basis for yield distribution and withdrawal entitlement. |
| **LTV** | Loan-to-Value ratio. Financing amount as a percentage of underlying collateral value. Platform targets conservative LTV ratios to protect provider capital. |
| **NTT** | Non-Transferable Token (Soulbound Token). Token that cannot be transferred between wallets. TradeNFTs are NTTs - they stay with the originating trade permanently. |
| **PEP** | Politically Exposed Person. Individual holding prominent public position. KYC screening includes PEP checks as part of enhanced due diligence requirements. |
| **Phytosanitary** | Certificate confirming agricultural products are free from pests and diseases. Required for cross-border food commodity shipments. Relevant to Food vertical pilot deals. |

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| **SBLC** | Standby Letter of Credit. Bank guarantee instrument that provides payment assurance if the buyer defaults. Preferred trade document for high-value deals. |
| **SCO** | Soft Corporate Offer. Initial offer from seller with pricing and terms, subject to negotiation. Early-stage document in trade deal formation. |
| **SGS Verax** | SGS (Societe Generale de Surveillance) inspection service. Independent third-party commodity inspection and verification. Used for dispute resolution re-inspection. |
| **STR** | Suspicious Transaction Report. Mandatory filing to UAE FIU (via GOAML) when transaction patterns suggest potential money laundering or terrorist financing. |
| **TCT** | Trade Collateral Token (Gold). ERC-20 token representing tokenized physical gold. 1 TCT = 1 troy ounce gold equivalent at LBMA spot price. |
| **UBO** | Ultimate Beneficial Owner. Individual(s) who ultimately own or control 25%+ of an entity. UBO declaration required during KYC for all entity registrations. |
| **UUPS** | Universal Upgradeable Proxy Standard (EIP-1822). Smart contract upgrade pattern where upgrade logic resides in implementation contract, not proxy. More gas-efficient than transparent proxy. |
| **VARA** | Virtual Assets Regulatory Authority. Dubai's regulatory body for virtual assets, established by Law No. 4 of 2022. Platform must obtain VARA VASP license before commercial operations. |
| **VASP** | Virtual Asset Service Provider. Entity providing virtual asset services (exchange, transfer, custody, etc.). Raw Essentials requires VASP license from VARA. |

# Appendix B: Document Version History

| Version | Date | Changes |
|---|---|---|
| **v1.0** | January 2026 | Initial SOW draft. Business concept, market analysis, high-level architecture. |
| **v2.0** | January 2026 | Implementation-level specifications. Added wallet architecture, oracle strategy, on-chain/off-chain boundaries, deposit/withdrawal flows. |
| **v3.0** | February 2026 | Complete scope consolidation. Added personas, 41 features, NFRs, operational scope, traceability matrix, module breakdown. |
| **v4.0** | February 2026 | Comprehensive revision addressing all 17 Engineering Manager review findings. Added: Assumptions & Constraints, Risk Register, Dispute Resolution Journey, Default Handling Framework, Liquidity Pool Mechanics, Acceptance Criteria for all 44 features, M5 Backend split into 4 sub-modules, Feature-to-Module Traceability, Delivery Roadmap with phases and milestones, Team Requirements, QA Scope, Change Control, Glossary. |

**DOCUMENT STATUS**

SOW v4.0 is the FROZEN BASELINE for Raw Essentials MVP Phase 0.

All 17 Engineering Manager review findings have been addressed in this version.

Any changes to this document must follow the Change Control Process (Section 20).

Next step: Stakeholder sign-off, then sprint planning begins.